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Scenario Evaluation of Malawi Energy Policy: LEAP System Modelling Projections from 2008 to 2063

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Scenario Evaluation of Malawi Energy Policy: LEAP System Modelling Projections from 2008 to 2063

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Abstract

National access to modern energy is deemed by Malawi government as the driving force to attain social-economic development. To achieve this, optimal decisions in energy planning and investment are considered important. This study evaluates potential pathways that will inform better policy design and investment option in energy sector. The Integrated Assessment Modelling method has been used to integrate social, economic and environmental variables that affect energy systems, and further analyses potential energy pathways in relation to these earth systems. The Long-range Alternatives Energy Planning system (LEAP) has been used to run a continuous time series from 2008 to 2063 of the three energy policy pathways (scenarios) using: (1) the current Malawi National Policy and Strategies on Energy (MwNEP) (2) the United Nation's Sustainable Energy for All strategy (SE4ALL) and (3) Business as Usual (REF) scenario. The analysis indicates an exponential growth in energy demand for both MwNEP and SE4ALL scenario with a slight higher demand in SE4ALL over MwNEP Scenario. In both cases Biomass remains an important energy source beyond the 2030. Thus, the SDG 7 will not be achieved by 2030. Similarly, energy demand trends in MwNEP and SE4ALL scenarios continue to grow beyond available energy stocks and generation capacity which will lead to a shortfall in energy supply. Environmentally, the GHGs emissions in MwNEP are higher than the SE4ALL scenario. This is reflected in the pro-environment policy objective pursued by SE4ALL scenario. In the medium-term LEAP analysis favours advancement of alternative renewable energies at household level like solar photovoltaic (PV) for lighting and sustainable biomass energy for cooking. In the long-run, scaling up the adoption of renewable energy in form of solar, wind and hydro-electricity to meet the energy gap. At the same time, designing and adhering to implementation of sustainable energy plans that cuts energy sources from fossil fuels and allow easy accesses to affordable renewable energies will result in a success of Malawi's energy system.

Keywords: Sustainable Development, Sustainable Energy, Integrated Assessment Model, Scenario framework, LEAP system, Malawi.

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Summary: Sustainable energy “indicates patterns of clean energy production and use that can support a society’s present and future needs with the least life-cycle economic, environmental, and social costs” (Yuzugullu, 2013). For Malawi, this can be regarded as the best direction that the country’s energy plan should be heading to curb the high energy insecurities along with outdated energy plans stipulated in the national energy policy of 2003. However, the path towards sustainable energy is chaotic and meshed with several factors. Among these are (1) social factors including population growth; (2) economic factors; (3) elapsing stocks of natural resource required to generate energy; and (4) energy transmission and consumption techniques. To evaluate the Malawi energy policy an Integrated Assessment Modeling (IAM) has been used because of its capability at compounding all these factors into a single analysis and generate potential sustainable energy system pathway. Kelly & Kolstad (1999) defines IAM as a multidisciplinary cross model that combines the natural sciences and economics in climate change issues, to evaluate policy options under climate change. This study uses three energy policy scenarios built from existing energy policies; (1) Malawi National Policy, and (MwNEP); (2) the United Nation’s Sustainable Energy for All strategy (SE4ALL); and (3) Business as Usual (REF) scenario. The LEAP (Long-range Energy Alternatives Planning) system, a tool that performs analysis in IAM, has simulated a 55year period energy demand and supply projections to showcase the trends in energy system of Malawi assuming either of the three scenarios were implemented. The analysis aims at deriving results that can contribute to long run energy planning for Malawi.

The study results show that Malawi energy demand trends correlates with the SE4ALL, with a slight higher modern energy (electricity) demand in SE4ALL plan over Malawi policy plan. This is attributed to national energy action plans that have incorporated SE4ALL strategies. At sector level, the household will continue to dominate the energy demand with biomass demand being higher than electricity. This suggests advocating for sustainable biomass energy in the medium term while the country transitions its electricity growth. This analysis indicates that by 2030, the electricity coverage proposed by SDG7 will not be achieved despite current progress towards pursuing the SDG 7 in Malawi. Furthermore, with increasing per capita income the demand for energy exceeds the current energy supply across the study period. Likewise, the demand for fossil energy from petroleum and coal fuels for transportation and energy generation give rise to GHG emissions. LEAP results favours energy policy planning towards decentralizing the production into off-grid and min-grid clean energy production from solar and hydrology at community level. This should cover more than 85% of unelectrified households and increase electricity supply. Additionally, in the medium term enhancing action plans towards sustainable biomass production to support household sector energy demands without harming the environment. Furthermore, importing grid energy from neighboring countries can cover any shortfall in the short term while pursuing long term objectives of total sufficient local energy production. Overall, the departure from the fossil energy dependence and incorporation of innovative clean energy development will enhance the country’s sustainable energy plans.

Keywords: Sustainable Development, Sustainable Energy, Integrated Assessment Model, Scenario framework, LEAP system, Malawi.

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List of Abbreviations

Government of Malawi

ADP	Malawi Agriculture Development Programme
BEST	Malawi National Biomass Energy Strategy
DoE	Department of Energy Affairs
FD	Forestry Department
GoM	Government of Malawi
MAREP	Malawi Rural Electrification Programme
MEGS	Malawi Economic Growth Strategy
MGDS	Malawi Growth Development Strategy
MNREM	Ministry of Natural Resources, Energy and Mining
MPRS	Malawi Poverty Reduction Strategy
MwNEP	Malawi National Energy Policy Scenario
NEAP	National Environmental Action Plan
NSREP	National Sustainable and Renewable Energy Programme
MAED	Malawi Energy Demand Report

Development partners

AfDB	African Development Bank
COMESA	Common Market for Eastern and Southern Africa
FAO	Food and Agricultural Organization of the United Nations
IEA	International Energy Agency
IRENA	International Renewable Energy Agency
SADC	Southern African Development Community
UN	United Nations
UNFCCC	United Nations Framework Conventions on Climate Change
USAID	United States of America International Development
VPoA	Vienna Programme of Action
WBG	World Bank Group

Other

ESCOM	Electricity Supply Corporation of Malawi Ltd.
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EGENCO	The Electricity Generation Company Limited
GDP	Gross Domestic Product
GHG	Greenhouse Gases
GHI	Global Horizontal Irradiation
HDI	Human Development Index (UNDP statistic)
ITCZ	The Intertropical Convergence Zone
LDCs	Least Developing Countries
LEAP	Long-range Energy Alternatives Planning
LPG	liquefied petroleum gas
LRMC	long-run marginal cost
LUCF	Land-Use change and forestry
MDGs	Millennium Development Goals
MK	Malawi Kwacha
SAPP	Southern African Power Pool
SE4ALL	Sustainable Energy for All Scenario
SDG	Sustainable Development Goals

Units of measurement

Emissions	MTCO ₂	Metric Tonnes of Carbon Dioxide Equivalent
Currency:	US\$	United States of America Dollars
	MK	Malawi Kwacha
Weight:	kg	kilogramme
	t	metric tonne (1,000 kg)
Volume	l	litre
Energy:	MJ	mega-joule (10 ⁶ joules)
	MJU	mega-joules of usable energy
	TJ	terra-joule (10 ¹² joules)
	Wh	Watt hour
	GWh	gigawatt hour
	TOE	Tonnes of Oil Equivalent
	w.e.	wood energy equivalent
	W	Watt
Time:	month	month
	yr	year

%

percent

1. Introduction

1.1. Background of the study

Human activities in the Anthropocene continue to shape the global environmental status (Steffen, et al., 2007). The activities pursued towards poverty eradication and global economic development overlap with the population growth and the increasing demand for goods and services, such as water, energy, and food. Moreover, it induces the pressure on the environment which pushes global resources beyond their carrying capacity threshold (Rockström, 2009). Therefore, to operate within the planetary boundaries while simultaneously promoting socio-economic wellbeing, countries enforce national and international institutions, and legal frameworks to ensure a sustainable world. The energy sector as a key social economic development sector has for years been the main attraction for more sustainability actions because of its vehement contribution to the Greenhouse Gases (GHGs) emissions that are pushing the biogeochemical flows and biosphere integrity beyond the global threshold (Rockström, 2009; Mandelli et al. 2014; IEA, 2017). To account for the future of the energy sector, long-term foresight projections on the energy demand and supply are required. The energy mix that has low environmental impact and the ability to meet the present demand without compromising future availability is deemed sustainable (IEA, 2017). To build such a case, this study will investigate Malawi, a country which struggles to secure sustainable economic development as a result of challenges in energy development and environmental protection (Gamula, et al. 2013).

This study builds distinct energy scenarios and emission factors using systems analysis simulations to investigate the energy demand, energy supply, and emission levels for a period span of 55-years (2008 to 2063). 2008 has been selected as a base year because of data availability as well as analysis level of significance when examining years between 2018 to 2030 which are critical to the Sustainable Development Goals (SDG) enforced by United Nations (UN). Moreover, in 2008 Malawi conducted more national studies including the *Annual Economic Report* and *population and household census*. Whereas, 2063 is not link to any statistical permutation rather it has been selected because it is the end year for the African Union's agenda 2063. African Agenda 2063 was formulated to guide Africa's development to 2063 (African Union Commission [AU], 2015 and 2016). The simulation results commentary on optimal energy mix has been supplanted by critical literature review to build optimal scenario cases for Malawi. Figure A9.1 in appendix shows the research outline.

1.2. Malawi Country Profile

1.2.1. Administration and Demographics

The Republic of Malawi consists of 28 districts spread across three regions: The south, central and northern region. Lilongwe is the capital city and three other commercial cities include, Blantyre, Mzuzu and Zomba. The size of residents population is increasing exponentially from 4.4 million in 1964 to 13 million (13,077,160) in 2008 population census, with an average growth rate of 2.3% (National Statistics Office, 2008). It is currently projected at 19.2 million people for 2017 (CIA, 2017). The share of urban and rural population census was 15.3% and 84.7% respectively in 2008 (GoM, 2008). Malawi's population is regarded as a young population with 46% of the population below the age of 15. This corresponds to the high dependency ratio of 83% (compared to 28% for Sweden) (GoM, 2016 and World Bank, 2017)



Fig. 1.1. Map of Malawi.

Source <http://www.map-library.com/map-library/maps-of-africa/maps-of-malawi/>

According to World Bank Group (2017) the age dependency ration for Malawi is a proportion of dependents (people below 15 years or above 64 year), to the working age population (ages between 15 and 64 years). This means that out of 100 persons in actively working population aged 15 to 64 years there are 85 persons (below 15 and above 64) who are younger or older and depends on them for their livelihood. Therefore, in the long term this dependency ratio increase creates non-productive population and shrinks the economic development of the country and contribute to extreme poverty levels (Pettinger, 2017). In addition, the country has

a relatively low economic investment in young adults in form of employment and conducive business environment (GoM, 2016). Per Malawi Labour Force Survey of 2014, 21% of the country's total labour force was unemployed (NSO, 2014). Youth (aged 15 – 24) unemployment was at 28% (NSO, 2014)

1.2.2. Government Structure

The government has three branches including the executive, legislature and judiciary branches. The executive has the power to make and implement policies and is comprised of the bureaucracy, cabinet ministries, departments, and sections (GoM, 2011a). The legislature comprises of the Members of Parliament (MPs) who act as legislators to approve bills and enact policies and assess the government budgetary plans (GoM, 2011a). The legislature has committees that represent different activities. For instance, the Parliamentary Committee on Natural Resources and Climate change act as a departmental committee on climate change environmental related issues. The Judiciary focuses on different legal issues of the country in conformity with the Malawi constitutional, and international standards. The Second National Communications to the Conference of the Parties (GoM, 2011a) expounds that, the judiciary is responsible for the constitutional standards, specialized statutes on environmental management for the betterment of all citizens. Regardless of the strategic governance structure, Malawi is currently ranked at 122 out of 180 countries in the global transparency index with an index score of 31 (Transparency International, 2017). This lower transparency score implies a high corruption level. Contrary, the country's corruption levels are far better than the lowest ranked Sub-Saharan countries like Somalia (9) and South Sudan (14), but worse than New Zealand and Denmark that are ranked as the least corrupt countries as the best countries with an index score of 89 and 88 respectively (Transparency International, 2017).

1.2.3. Social Economic Development

Malawi is classified as a Least Developing Country (LDC) with a Human Development Index (HDI) of 0.476 in 2015 (UN, 2018). In 2008 the country registered a robust growth in real GDP of 9.7% which was above the 6% average for Sub-Saharan African Countries (GoM, 2009). The national Gross Domestic Product (GDP) and GDP per capita for 2008 were at United States Dollar (US\$) 3.67 billion and US\$270 respectively (GoM 2009; and 2017b). Besides the small economic base, increasing population growth, young and heavily dependent population, and land locked status, the country has a good agriculture-based economy. Approximately 34.1% of the GDP comes from the Agriculture sector, 15.4% and 50.5% from industry (manufacturing, construction, energy, and mining) and service respectively (GoM, 2011a and 2016).

The agriculture sector is predominately subsistence rain-fed agriculture, and the sector operates with an under-developed technical and financial infrastructure (GoM, 2016). Approximately 90% of agricultural value-addition comes from the 2 million subsistence smallholder farm families (FAO and GoM, 2015). There are no concise figures on total irrigated land versus rain fed agriculture for Malawi (FAO and GoM, 2015). However, the National Investment Profile on water for agriculture and energy report contends that, all the smallholder farming depends on perennial rainfall and less than 20% of the potential irrigation land is being developed (FAO & GoM, 2015). This makes the agriculture sector highly vulnerable to the climatic shocks presented by climate change. In other words, Malawi being an agro-based economy, any climatic shocks can inevitably affect agricultural output leading to an economic mishap. The emergent of the 2017 Malawi Growth Development Strategy (MGDS) III designed for the 5-year period emphasizes on the importance of energy development as the “lifeblood of the

economy” that can juxtapose all other economic and social services in Malawi (GoM, 2017a). In summary, Malawi’s economic plans have been designed to push for transformation of Malawi into a middle-income country by 2020 (GoM, 1999).

1.2.4. Malawi National Strategies

The Vision 2020 is a parent national strategy on which all other strategies are built on alongside other regional and international strategies. Malawi launched Vision 2020 in the year 2000 as a 2020-year long-term sustainable national development policy with the aim of becoming a middle-income country. Vision 2020 is supported by national medium-term policies, sectoral strategic frameworks, and international strategies.

For instance, locally, the following policies were enacted: the Malawi Poverty Reduction Strategy (MPRSP) of 2002, Malawi Economic Growth Strategy (MEGS) of 2004, Malawi Growth and Development Strategy (MGDS) of 2006, National Environmental Action Plan (NEAP) of 1994 and revised in 2002, National Environmental Policy of 1996. Currently, the Malawi Agriculture Development Programme (ADP) of 2007 that focuses on agricultural components of the MGDS; Malawi Growth and Development Strategy (MGDS) of 2011 to 2016, that supported the work of Millennium Development Goals (MDGs); and the Malawi Growth and Development Strategy (MGDS) of 2017. Internationally, Malawi abides by different strategies. Subsequently the most relevant for discussion in this study are the Rio convention of 1992, Sustainable Development Goals (SDGs) of 2015, Kyoto Protocol of 2002, Paris Agreement 2015, Istanbul Program of Action of 2011, Vienna Programme of Action (VPoA) of 2014, Agenda 2063 of 2013, Southern African Development Community (SADC) Regional Indicative Strategic Development Plan of 2003, and the Common Market for Eastern and Southern Africa (COMESA) Treaty of 1994. All these instruments have been installed to cater for the nation’s development plans.

The process of transforming Malawi into a middle-income country economy is reinforced by the Vision 2020 in three ways. Firstly, the Vision 2020 aims at developing the agricultural sector for domestic and export demand (GoM, 2016). Secondly, enhancing sustainable water resource use and management (GoM, 2005). Lastly, transitioning from a biomass energy sector into a modern energy component. Figure 1.4 below has summarized these processes (GoM, 2003a). The inextricable water, energy and food link outlined here underpin Malawi’s development progress (GoM, 2003b). Moreover, to attain a sustainable future requires sustainable energy. Sustainable energy “indicates patterns of energy production and use that can support a society’s present and future needs with the least life-cycle economic, environmental, and social costs” (Yuzugullu, 2013). Sustainable energy does not include any sources that are derived from fossil fuels or waste products, rather they are replenishable and helps reduce the greenhouse gases (GHGs) (UN, 2011; Yuzugullu, 2013). Sustainable energy accounts for the interrelations with other sectors like industry, agriculture, environment, climate change, health, transport, and Information Communication Technology. It further abides by the proposition of the United Nation’s Sustainable Development Goal (SDG) number 7, “by 2030, ensure access to affordable, reliable, sustainable and modern energy for all” (United Nations, 2016). To sum up, Malawi’s Social Economic development is driven by multiple strategies that incline towards building an economically resilient Malawi.

1.3. Malawi’s Energy System

Malawi is relatively well endowed with energy resources in biomass, biogas, coal, perennial rivers for power generation, and adequate solar irradiation for photovoltaic and other solar applications; as well for nuclear energy sources (Suri et al., 2015 and GoM, 2016). However,

the potential for these resources is far from being fully realized (GoM, 2003a). In 2008 Malawi's main energy mix comprised of 88.2% biomass called "traditional fuels" (wood – 52%, charcoal – 41%, and residues – 7%), 6.79% Petroleum products, 2.5% hydro-electricity, 2.39% coal, and 0.06% other alternative renewable energy (GoM, 2011b). Figure 1.2 below summarizes the total primary energy demand by sector which was at 4,125.97 kTOE in 2008 (GoM, 2011b).

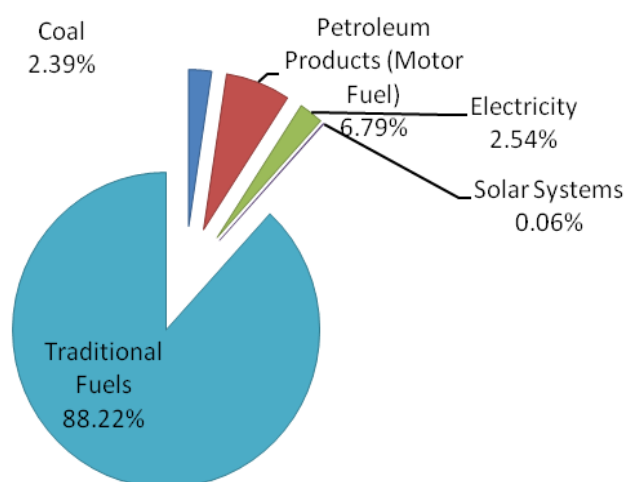


Fig. 1.2. Malawi Total Energy Demand Share 2008
Source: Malawi Energy Demand Report (GoM, 2011b)

The renewable sources include solar systems, biogas, wind energy, and mini- and micro-hydro systems (GoM, 2012). Biomass (traditional fuels) is considered a conditionally renewable source of energy, though current practice in Malawi is to refer only to wind, hydro and solar PV as "renewables" (GoM, 2009). However, for this study, hydroelectric-power is being referred to as Electricity. More than 90% of the population uses Biomass in form of charcoal and wood (GoM, 2017b). Likewise, access to electricity has grown from around 9% in 2008 to 12% of the population in 2017 (NSO, 2009; GoM, 2017b; USAID, 2017). In the base year the actual peak load was 242 MW but currently, it is estimated at 600MWh of electricity energy demand (GoM, 2011b and 2017) in comparison to Sweden's 2016 energy demand of 370 TWh (Swedish Energy Agency, 2017).

Moreover, conventional energy systems described here, contribute decisively to global environmental degradation, and increasing GHG emissions contribution to the planetary boundary threshold (GoM, 2011a). To avert this burden, proper policy planning that accounts for the targets of the seventh SDG can ensure a sustainable energy future (Hayes and Schofield, 2002). Thus, there is a necessity for designing energy systems that are increasingly more efficient, less expensive and cleaner. The Malawi National Energy Policy (2003) adheres to the following three long-term goals (GoM, 2003):

- make the energy sector sufficiently robust and efficient to support GoM's socio-economic agenda of poverty reduction, sustainable economic development, and enhanced labour productivity;
- catalyse the establishment of a more liberalised, private sector-driven energy supply industry in which pricing will reflect the competition and efficiency that will have developed through the reform process;
- transform the country's energy economy from one that is overly dependent on biomass (93%) to one with a high modern energy component in its energy mix. A biomass -

commercial energy (hydro-electricity) mix target of 50% - 50% is set for 2020. The year 2020 can be compounded into the Energy Investment Plan end year of 2030.

To support the success of the main goals of the Malawi National Energy policy, the country developed the following energy strategies and legal instruments:

Table 1.1. Malawi Energy Legal Instruments and Strategies

Strategies	Legal Instruments
National Sustainable and Renewable Energy Programme (NSREP) of 1999	Electricity Act (2004) and the amended (2016)
MGDS II and Current MGDS III (2011 and 2017)	Liquids Fuels and Gas Act (2004)
Malawi National Biomass Energy Strategy (BEST) 2009	Energy Regulation Act (2004)
Malawi Electricity Investment plan 2010	Malawi Electricity Act 2004
Malawi Rural Electrification Programme (MAREP)	Rural Electrification Act (2004)
Independent Power Producer (IPP) Framework for Malawi 2016	
Malawi National Cook stoves Roadmap 2016	

Clearly, from table 1.1 above it shows that the country's energy focus on increasing household access to modern energy (electricity). Moreover, households have the highest energy consumption share (84%) dominated by biomass. However, encouraging modern energy growth will tremendously reduce biomass dependence. This will inevitably save time spent fetching wood for other socio-economic development tasks (IEA, 2017), while at the same time reducing the risk of acute respiratory infections (ARI) coming from cooking in smoke and increase carbon sequestration due to reduced fuelwood and charcoal use (Po et al. 2011). These factors correspond to the energy policy forecasts of transforming the energy mix from the high biomass dependence to modern energy consumption.

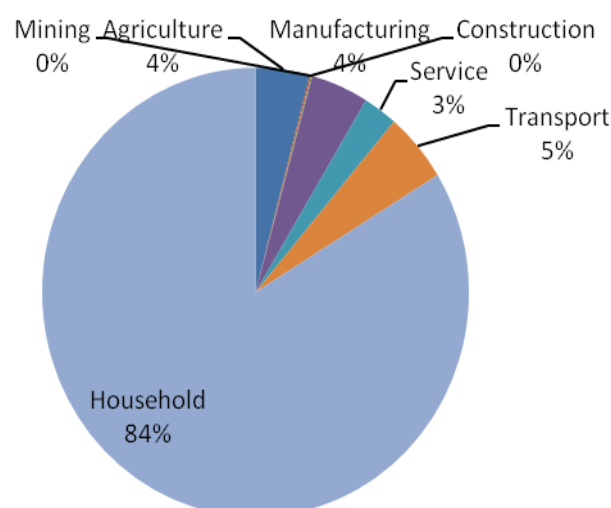


Fig 1.3. Total Sectoral Energy Demand (at 4,125.97 kTOE) in 2008 (GoM, 2011b)
Source: Malawi Energy Demand Report (GoM, 2011b)

1.3.1. Energy Demand and Supply

According to the *Integrated Household Survey* (GoM, 2017b), the energy use at the household level is highly influenced by the urban-rural population dispersion; where around 99% rural households use biomass as the main primary energy and about 90% of urban rely on biomass. This is attributed to a low rural electrification which is at 3% in comparison to the 43% electrification in urban areas (GoM, 2017). On the other hand, electricity demand continues to grow at 6-8% per annum (GoM, 2010). Consequently, the current system is strained making the electricity sector unreliable and prone to numerous power outages (Taulo et al. 2015; NSO, 2013). Despite variation in electrification, both rural and urban areas have similar energy consumption behaviors for cooking.

Although the biomass has a high consumption levels (3640 kTOE) and that it has a well-defined strategy, it remains as one of the most unsustainable energy sources (GoM 2018). For instance, biomass energy source like forests continue to be cleared for agriculture and charcoal production without reforestation. Between the year 1991 and 2008, about 669,000 hectares of forests cover were converted to agriculture and 70 million cubic metres of wood were cleared for agriculture expansion (GoM, 2009b). Moreover, in 2008 alone the area cleared was estimated at 45,000 hectares (GoM, 2009b). Regardless, the traditional three-stone fuelwood cooking process dominates most households, especially in the rural areas. This contributes to excessive wood usage, carbon monoxide (smoke) emissions during the cooking process, that contributes to Household Air Pollution respiratory diseases (Rylance, et al., 2016; Langbein, 2017; Das et al., 2017).

As asserted above, electrification rates (9% - 2008 and 11% - 2016) in Malawi are among the lowest in sub-Saharan Africa. As of 2008, the total installed capacity was around 360MW (372.64 MW as of 2016) is being supplied by 350.94 MW from hydropower plants and 21.7 MW from standby diesel power plants (GoM 2017b). Table 1.2 below summarizes the total energy demand for Malawi in base year 2008.

Table 1.2. Total Energy Demand for Malawi in Base Year 2008 (kTOE)
Source: GoM, 2011b

Economic Sectors	Kerosene	Motor Fuels		Coal	Electricity		Solar Systems	Biomass (Traditional Fuels)	Total
		Petroleum Products (Gasoline & Diesel)	Ethanol		Specific ¹	non-specific ²			
Agriculture	-	49.5		1.5	6.30	0.70	-	110.00	168.0
Construction	-	5.00			-	-	-	-	5.00
Mining	-	0.8			2.85	0.15	-	-	3.8
Manufacturing	-	2.00		96.19	25.30	6.73	-	50.00	180.22
Services	1.00	-		0.99	13.19	10.30	1.10	80.00	106.58
Transportation	-	198.45	8.68		-	-	-	-	207.13
Household	14.64	-			18.39	20.91	1.30	3,400.00	3,455.24
Total	15.64	255.75	8.68	98.68	66.03	38.79	2.40	3,640.00	4,125.97

¹Specific use of electricity refers to those electricity uses that are non-substitutable such as lighting and powering electric appliances.

²Non-specific use of electricity refers to those electricity uses that can be substituted such as heating, processing and cooking.

Despite this low electricity rates, the Malawi government through the Independent Power Producer (IPP) Framework (GoM, 2017b) encourages other corporate players to enter the electricity production sector. This is the supplement to long-term plans of constructing 300MW coal power plant by early 2020s (GoM, 2017b). The newly established government parastatal, Electricity Generation Company of Malawi (EGENCO, 2018a) has been commissioned to rehabilitate and upgrade the Nkula and kapichira hydro power station (EGENCO, 2018a). In addition, the company has aligned new projects to build a 20 MW Solar PV plant. Furthermore, projects are underway for the installation of 36 MW diesel generators across the three major cities of Malawi, Blantyre (Mapanga - 20MW), Lilongwe (10 MW) and Mzuzu (6 MW) (EGENCO, 2018a). Currently the Mapanga electricity generators are expected to be in operation by June 2018 (EGENCO, 2018b). The aim is to increase installed electricity capacity above 719 MW by 2020 and increase access to electricity to around 30% of the population by 2030 (GoM, 2017b).

The absence of a refinery system in Malawi, means that the country imports 100% of its petroleum products (gasoline and diesel inclusive). The ethanol that is produced locally at industrial level is blended with gasoline in a ration 90:10 (gasoline:ethanol) for consumption (GoM, 2009b).

Malawi's GHG emission is regarded as one of the lowest at global level (Global Carbon Atlas, 2016), emitting 1.3 MtCO₂ equivalent in comparison to 10,151 MtCO₂ equivalent for China (UNFCCC, 2015; Global Carbon Atlas, 2016). Regardless of small figures of carbon dioxide equivalent emission, it still contributes to the accumulation of carbon in the earth system that is contributing to the global warming. At national level the GHG inventory in the UNFCCC's National Communications (GoM, 2011a) reported that, Land-Use change and forestry (LUCF) have the most significant GHG emission levels accounting to 56% of the emissions. The second highest emission came from agriculture at 40%, and the energy sector was at 3%.

1.4. Sustainable Energy for All (SE4All) and Sustainable Development Goal (SDG) 7

In 2012, the UN General Assembly introduced the Sustainable Energy for All (SE4All) initiative as a reaction to the urgent energy challenges and environmental degradation caused by traditional energy resources (United Nations, 2011). The initiative spearheaded by the high-Level Group on SE4All, seeks to identify and mobilize action by all stakeholders in support for energy access, energy efficiency, and increasing renewable energy. The initiative is enshrined in the Sustainable Development Goal number 7 “Ensure access to affordable, reliable, sustainable and modern energy for all” (UN, 2015). The SE4All objective on “*universal access to modern energy services*” corresponds to the SDG7 indicator 7.1 “By 2030, ensure universal access to affordable, reliable and modern energy services”. Furthermore, the SE4All objective on “*improvement in energy efficiency*” ties with SDG indicator 7.2, “By 2030, increase substantially the share of renewable energy in the global energy mix”. Lastly, the objective on “*doubling the share of renewable energy in the global energy mix*” corresponds to the indicator 7.3 “By 2030, double the global rate of improvement in energy efficiency” (UN, 2015).

Recognizing the energy challenges, Malawi along with 43 other African Countries joined the initiative. However, according to the latest SE4All African Hub Annual report (AfDB, 2017), Malawi was categorized as one of the member countries which is at the development phase of implementation. Thus, understanding the global initiative energy trend laid forth in the SE4All objectives and further contextualizing them into Malawi terms can help project long term energy plan comparison to the existing energy plans.

1.5. Problem Statement

The main energy sources in Malawi comprises of biomass (traditional fuels), hydro-electricity, coal, imported petroleum and solar (See figure 1.2). However, only two energy sectors have well defined strategies, the hydro-electricity and biomass sector (Gamula, et al, 2013). The Hydro-electricity is driven by the Malawi Electricity Investment plan of 2010, while the Malawi Biomass Energy Strategy of 2009 for biomass. Although Uranium has potential for nuclear energy, but the potential is unlikely to be developed because the nuclear technology and infrastructure education necessary to build and operate a nuclear power station is not available. As such it is currently being mined for export only.

The dilemma of pursuing energy strategy and the consecutive energy demand posed now and that which is foreseen in the near future coming from the growing youth (under 15-years) population who make up 46% of population, present two scenarios that can be juxtaposed. On the one hand, the energy policy was formulated to reflect the vision 2020’s objectives. Despite the presence of constitutional energy instruments, the energy policy has not been updated since its launch in 2003 until recently in 2018 when the government initiated a new revised energy policy which is yet to be enacted and published. As the year 2020 is approaching the policy requires further enhancement or a complete change of direction to meet the national long-term sustainable energy future beyond 2020.

Although no comprehensive policy has been enacted to ensure energy security after 2020, medium-term and short-term energy reforms have been implemented which extend beyond 2020. For example, the MDGS III, the rural electrification program, Malawi Electricity Act, Renewable Energy Targets. Nevertheless, since 2015 no legislative committee specializes directly on energy issues like the former Parliamentary Committee on Natural Resources

energy and mining has been installed in the parliament. Consequently, there is a failure threat on discussions related to energy issues by the policy makers.

On the other hand, Malawi seeks to build a sustainable energy future (GoM 2003b). However, the energy policy outlines conflicting objectives in its future energy plan. Figure 1.4 summarizes the energy projection plan in the Malawi National Energy Plan of which fossil energy (coal) is expected to increase. In addition, the plans to install diesel electricity generation systems will be contrary to the idea of pursuing clean energy production. Besides the focus on building a sustainable energy, the young population is growing up in a world with high energy demand for technology, mobility, cooling, and other functions. As a result, a well-rounded energy system required going forward will demand sustainable energy.

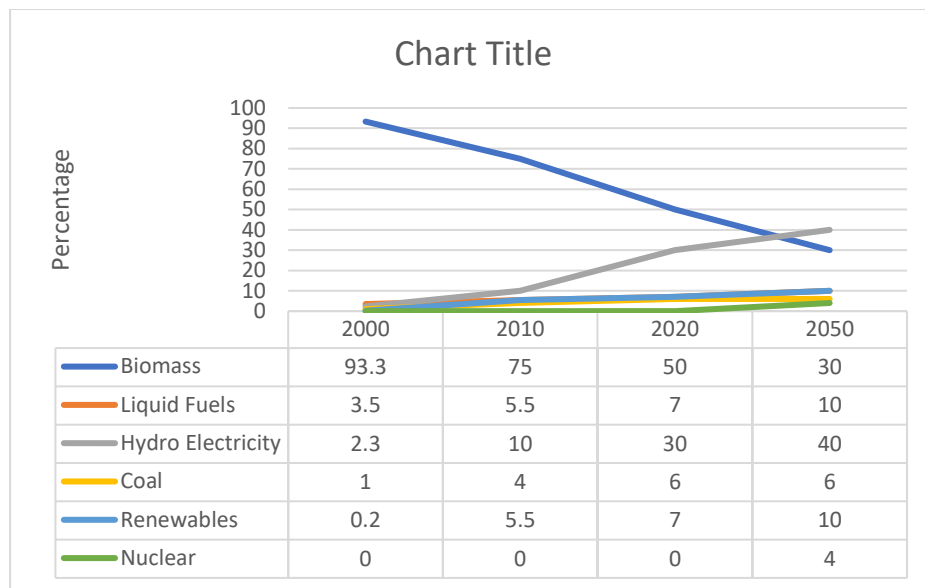


Fig. 1.4. Malawi Energy Demand Shares Projections. Source: GoM, 2003a

In a recent study on the current energy situation of Africa, Mandelli et al. (2014) use up to date data, to describe the concept of sustainable energy development, and see if and how energy policies promoted by local players fit within the sustainable energy pathway. However, Malawi is not included in this comprehensive assessment study. This presents gaps in recent studies on sustainable energy planning for Malawi (Kaunda, 2013). Yet, there is a demand for more specific studies that investigate the energy policy side to inform decision makers (Kaunda, 2013). Moreover, there are no scenario-based studies that project Malawi's energy system beyond 2050 while comparing the national energy plan (MwNEP) and the sustainable energy for all (SE4ALL). This project aims to contribute to the energy sector by informing the existing research gaps. It does this by assessing future energy projection and emission level for both National Energy plans and the plans stipulated in the SE4ALL goals. To ascertain the future projections this study employs a state-of-the-art modelling system called Long-range Energy Alternatives Planning System (LEAP) developed by Stockholm Environmental Institute.

1.5.1. Aim and Objective

The aim of this project is to assess the sustainable energy policy planning for Malawi. This is achieved through review of pertinent literature, and scenario modeling in LEAP simulation tool. Two main objectives are pursued in this project. Firstly, a critical policy analysis to establish the relationship between Malawi strategic documents on energy and sustainable

energy futures. Secondly, to compile data and use the LEAP system to project energy demand between 2008 and 2063 and simulate carbon intensities from the Intergovernmental Panel on Climate Change (IPCC). These findings will be adopted to simulate a state-of-the-art assessment scenario, to find the compatibilities of the current energy policy compared to futuristic alternative sustainable energy scenario in line with SDG 7.

1.5.2. Research questions

The following questions shall be addressed:

- How are the energy demand trends in existing Malawi energy policies compatible with sustainable energy as specified in SE4All and the SDG 7?
- What are the relationships between projected future energy demand and available current energy resource stocks?
- What alternative energy planning, energy source and supply options can be leveraged to attain a sustainable energy future, and reduce emissions?

2. Methodology

2.1. Systematic Literature Review

This study employs a systematic literature review on Malawi energy literature to provide a reliable, scientific overview of extant research on the study. The core aim of using this approach is to identify, appraise and synthesize all relevant studies using a transparent, replicable process (Tranfield et al. 2003 and Petticrew & Roberts 2006). The study systematically selects literature and quantitative data that directly relates to the research questions under study using seven stages of carrying out a systematic review developed by Petticrew and Roberts (2006). Thus, the focus is only on data that is relevant to the analysis of this study. This method is important because there are few systematic reviews on Malawi energy. Likewise, the method avoids information overload from many closely related sources. To apply this method, the study developed a schema (see figure 2.1 and table A9.1 in appendix) based on Petticrew and Roberts (2006). This shows how the literature and data were systematically identified appraise synthesized and reviewed in this study. The research questions and data required were categorized in the first box labeled Research Question (RQ). The second phase labeled “source” represents the types of studies that need to be located in order to answer the research questions. See section 2.5 on data source. The third sections correspond to a comprehensive literature search that is identified in relation to the question in the RQ phase. The fourth stage involves screening the results of the studies in phase 3 and deciding the ones that meet the inclusion criteria. This involves a more detailed examination. All the literature and data were later appraised in “phase five” through assessment of every literature and data set to determine its quality. The last two phases involved assembling and synthesizing data and its corresponding heterogeneity among the study findings. Finally, the literature and data are disseminated in this study. Table A9.1 in the appendix presents the procedure employed for this process.

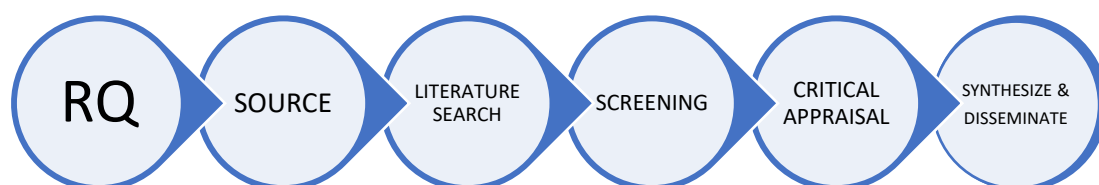


Fig. 2.1. Systematic Literature Review Schema

2.2. Modelling

Integrated Assessment Modelling (IAM) approach has been applied in this study because it has the capability to assess and project the interactions of key features of human system (like the social demographics, energy use techniques, energy technologies, economy, transport, etc.) and the environmental processes (like climate change and its impacts) (Moss, 2010). AIM is a multidisciplinary cross model that combines the natural sciences and economics in climate change issues, to evaluate policy options under climate change (Kelly & Kolstad, 1999). It has been used to draw scenarios (see section 4.2.1) that best describe Malawi energy system under different policy strategies. To apply the modelling approach an IAM tool developed by Stockholm Environmental Institute called LEAP (long-Range Energy Alternative Planning

System) has been used to simulate energy projections from 2018 to 2063 for different policy conditions. LEAP is a software tool used for energy policy analysis to track energy consumption and production, resource extraction, and accounting GHGs emissions (Heaps, 2012).

There are several types of tools for long-term energy forecasting which can be divided into seven categories, including simulation (e.g., RAMSES, BALMOREL, LEAP, WASP, etc.), scenario (e.g., MARKAL/TIMES, MESSAGE, LEAP, etc.), equilibrium (e.g., MARKAL, PRIMES, etc.), top-down (ENPEP-BALANCE, LEAP, etc.), bottom-up (HOMER, RAMSES, MARKAL/TIMES, MESSAGE, etc.), operation optimisation (BALMOREL, MESSAGE, RAMSES, etc.), and investment optimisation tools (MESSAGE, MARKAL/TIMES, RETScreen, etc) (Ouedraogo, 2017).

Despite this array of tools, the Long-range Energy Alternatives Planning System (LEAP) developed by Stockholm Environmental Institute (SEI) has excellent qualities at integrated, scenario-based modelling. LEAP stands out as an appropriate tool for this project. First, it is based on scenario analysis which is the focus of this project. Second, it requires limited initial data input which is conducive for this project which has high data limitations. Besides, LEAP provides a Technology and Environmental Database (TED) which offers up-to-date energy technologies data from a wide range of sources (SEI, 2008). Fourthly, LEAP system offers data visualization for end users which makes it easy for the researchers and policy makers to interpret the data without requiring an expert explanation. Fifthly, LEAP can track energy consumption, production and resource extraction in all sectors of an economy. Additionally, it has the capacity to develop scenarios in an interactive visual platform of its application (SEI, 2017). Consequently, LEAP tool serves as the optimal system to understand the long-term energy projection of Malawi's sustainable energy future.

The main components of the LEAP tool in this study are population size, economic growth (GDP), strategic government plans on energy efficiency and improvements, and final energy demand (fuel use by sector in TOE, final energy intensity per devices, technology cost and performance). In the LEAP system, data structures are arranged in hierarchy (tree) to model specific energy system. In this study the LEAP tree presented in figure 2.3 below has five end use energy demand branches. These include household, service, transport, agriculture and industry. In addition, the Key assumptions which have been described below (section 4.2), have been specified in the building of this model under the key assumption branch. Lastly, the transformation branch is composed of energy sources and transmission.

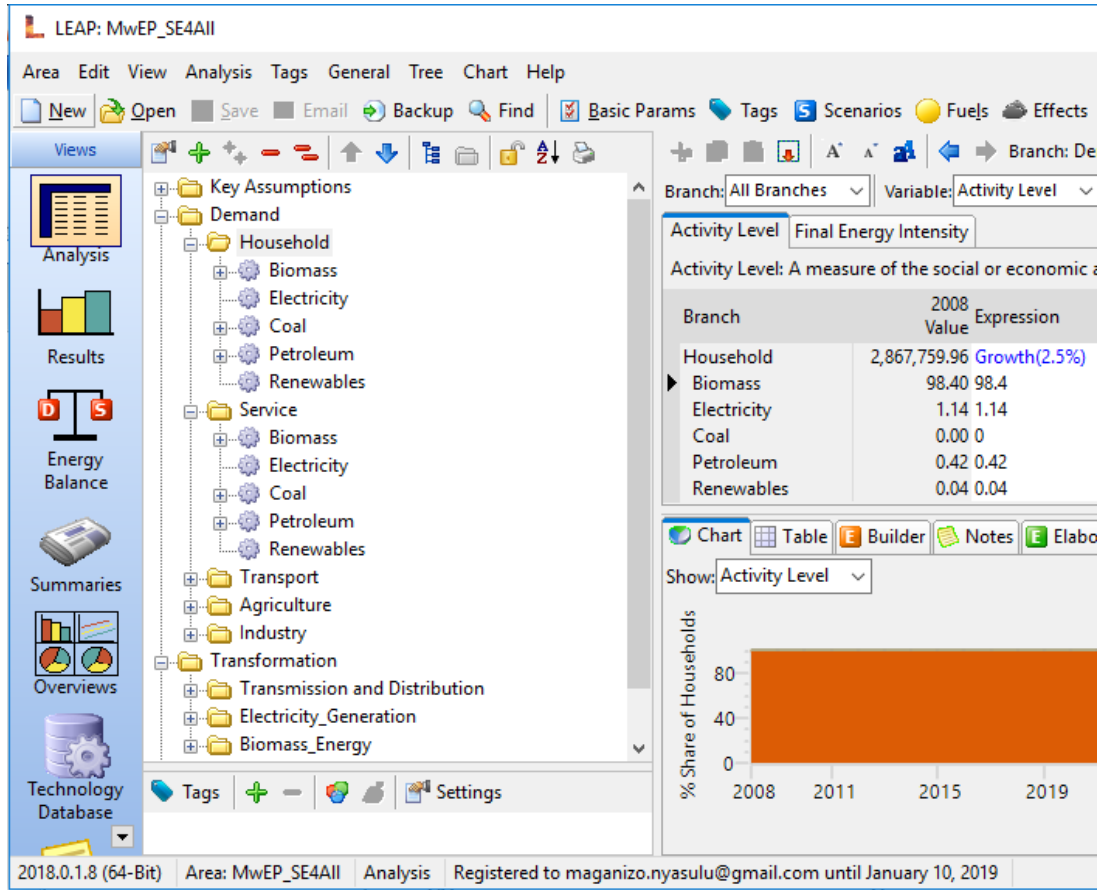


Fig. 2.2. LEAP Tool Screenshot

This study has compiled data through systematic literature review (see section 2.2 above) and simulated it in the LEAP system. The three scenarios conditions in this study are; (1) Business as Usual (REF); (2) Malawi National Energy Policies and Strategies (MwNEP) and (3) Sustainable Energy for All (SE4ALL) which also encompass Sustainable Development Goal number 7. The results presented energy demand (in TOE), supply and emission contribution. Figure 2.4 below summarizes the data process in the LEAP model.

2.2.1 LEAP Algorithm

The energy demand and GHGs emissions inside the LEAP model have been calculated using the following equation processes adapted from studies by Heaps (2012) and Ouedraogo (2017).

Energy demand

$$ED_{b,s,t} = AL_{b,s,t,y,p} \times EI_{b,s,t,n} \quad (1)$$

Where ED is the Malawi Total Energy Demand, b is demand branch (household, industry, agriculture, etc.) s is scenario, t is year. AL is the Activity level disaggregated by GDP average national growth rate (y), household population (p). The EI is the energy intensity per unit of activity calculated as;

$$EI_{b,s,t} = \frac{EC_{b,s,t}}{TC_{b,s,t}} \quad (2)$$

where EC is the electricity consumed, TC is the total consumers, b is demand branch; s is the sector; and t is the time. From Equations (1) and (2), the energy demand is forecasted for each

end-user. The net energy consumption for transformation has been calculated inside the LEAP system using the following equation:

$$ET_p = ETP_{sec,tec} \left(\frac{1}{f_{p,sec,tec}} - 1 \right) \quad (3)$$

ET is the net consumption for transformation (energy loss for transformation process) ETP is the product from the transformation process, f is energy transformation efficiency, tec is technology, p is the type of primary energy, and sec , is the type of secondary energy.

Carbon emissions

The carbon emissions from final energy consumption (CE) and the carbon emissions from energy transformation (CET) are calculated as follows:

$$CE_{p,s,t} = TA_{b,s,t} \times EI_{b,s,t} \times EF_{b,s,t} \quad (4)$$

$$CET = ETP_{sec,tec} \times \frac{1}{f_{p,sec,tec}} \times EF_{p,sec,tec} \quad (5)$$

$CE_{p,s,t}$ is carbon emissions, $EF_{b,s,t}$ is the carbon emission factor from sector or branch (b) scenario (s) and year (t). TE and EI are as specified above. CET is carbon emission from energy transformation, EF is the emission factor from one unit of primary fuel type (p) used to produce the secondary fuel type (sec) through the technology (tec) and ETP and f as specified above

Figure 2.4 below summarizes the LEAP process. Firstly, assumptions for each scenario were set in line with the timeline, social and economic activity levels (see chapter 4). This data on different activity level, social or economic activity for which energy is consumed are later coded into the system. Correspondingly, data on annual final consumption of energy per unit of the activity set in the first step were coded. This data is called final energy intensity. The emission factors were coded using the IPCC tier 1 module that come with the LEAP Technology Database (TED) tool in the LEAP. The analysis results were simulated as demand projection, emission levels and energy transformation.

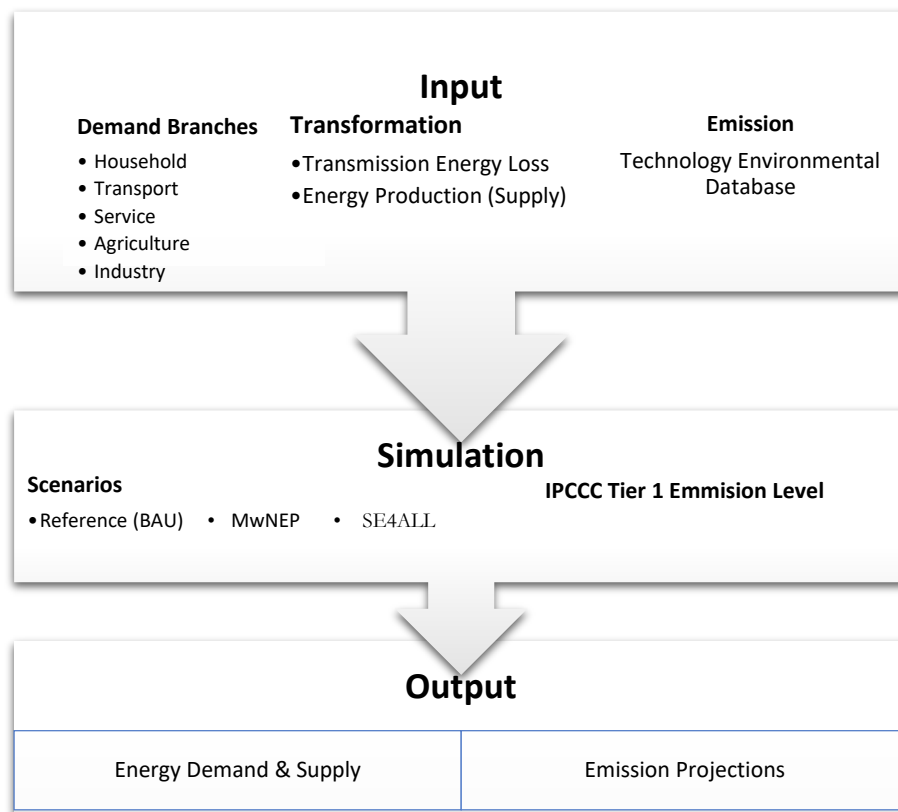


Fig 2.3. Data Process in LEAP

2.3. Data Source

To achieve the set research questions, data has been sourced from three categories; (i) Malawi data, (ii) UN data, and (iii) Other research literature studies that were appraised as relevant to the study objectives. The study has used the Malawi National strategies that discusses energy plans and National Energy Policy, National demographics, and economic statistics.

Malawi Energy Statistic Data: Energy related data in Malawi is collected through censuses, surveys and as administrative data. Key literature data sources for Malawi data include the Department of Energy affairs in the Ministry of Natural Resources, Energy and Mining, and National Statistics Office (NSO). Most of the energy data has been compiled from Population and Housing Census reports (NSO, 1999 & 2009), a series of Welfare Monitoring Survey, Multiple Indicator Cluster Survey, Malawi Demographics Health Survey, Integrated Household Survey (2, 3 and 4) and other energy reform strategies and survey reports. Furthermore, this study used the activity level data, and energy balance data provided by the Department of Energy Affairs (GoM, 2011b), ESCOM energy consumption figures (ESCOM, 2011), Biomass consumption figures from BEST (2009) and research literatures (Gamula et al., 2013).

UN Data: The second source of data has been from the United Nations documents. These include World Bank Group data (2018), World Human Development index, UN Energy reports and UNFCCC national communications on Malawi and Sub-Saharan Africa, and National Reports to UNFCCC. In addition, data from literature that directly links to the study objectives.

Other Literature: To supplement the data and literature, the study has used search engines provided by the Uppsala University Library (www.ub.uu.se), Nordic African Institute Library, Swedish University of Agricultural Sciences (SLU) library, Elsevier's Science Direct, JSTOR. Other sources include the International Renewable Energy Agency (IRENA), and International Energy Agency (IEA).

2.4. Study Limitations

The study focuses on secondary data and literature reviews. This approach is potentially limiting because it provides only sectoral or regional specific analyses, especially on energy consumption and demand for unsolicited sectors/services. This limitation concurs to Ouedraogo's (2017) observation on most sub-Saharan African countries where "data on traditional energy prices and supply, and on informal sectors are not always available". However, due to time and financial constraints to run a primary survey, the study employs a LEAP modeling system that require less data to draw simulation outputs. Despite these LEAP specifications, Malawi government data sets did not provide all the required data per sector (activity level and final energy intensity data), as such the study complemented the Malawi data described in section 2.3 above with aggregates from other countries in Sub-Saharan Africa with energy systems similar to Malawi like Zambia, Namibia, Mozambique, Lesotho and Burundi. At the same time, the study has reviewed official literature and datasets from legitimate stakeholder (IEA, IRENA, UN world databank, etc.) to compound on data reliability. Furthermore, different policies are under review, for instance the Malawi National Energy Policy, as such most of the analysis in this study has been based on the 2003 National Energy Policy but with a keen eye on the drafted Malawi National Energy Policy of 2018.

With exception of the petroleum energy products which are 100% imported, this study assumes that other energy sources like electricity, biomass and renewables are sourced locally. This is in line with less energy imported from bordering countries.

3. Scenario Framework

Chapter 3 outlines a descriptive process of the conceptual framework employed for the analysis of this study. The scenario framework approach that has been employed in this study, has been defined by Moss et al. (2010) and Ebi et. al. (2014) as a new interdisciplinary research assessment framework that explores a range of probable future climate and related physical changes that could pose risk to human and natural systems.

3.1. Scenario Framework Approach

Integrated assessment models represent key features of human systems, such as demography, energy use, technology, the economy, agriculture, forestry and land use (Moss, 2010). The models are used to develop emissions scenarios, estimate the potential economic impacts of climate change and the costs and benefits of mitigation, simulate feedbacks, and evaluate uncertainties. These scenarios provide plausible descriptions of how the future might unfold in several key areas—socioeconomic, technological and environmental conditions, emissions of greenhouse gases and aerosols, and climate (Moss et al, 2010; van Vuuren et al. 2014).

This study uses the parallel scenario approach which is ‘representative’, in that each Representative Concentration Pathways (RCP) provides only one of many possible future scenarios that would lead to the specific radiative forcing characteristics (Moss et al., 2010). This approach reduces the magnitude of patterns to a specificity of the scenario. Besides, RCPs are potential pathways that are based on the agreed conditions (including conditions constructed from literature review) and are considered in building scenarios (Moss, 2010).

This parallel scenario process proposes four Representative Concentration Pathways. The RCP 8.5, RCP6.0, RCP4.5, and RCP2.6. Simulations are built based on the variables (radiative force, carbon concentration, and reference factor) as provided by the pathways below (Moss et al. 2010).

RCP8.5: Radiative force of $>8.5 \text{ W m}^{-2}$ in 2100, and Carbon concentration (p.p.m) $>1370 \text{ CO}_2$ equivalent.

RCP6.0: Approximate 6 W m^{-2} radiative force at stabilization after 2100, and carbon Concentration (p.p.m) around 850 CO_2 equivalent.

RCP4.5: Approximate of 4 W m^{-2} radiative force at stabilization after 2100, and carbon Concentration (p.p.m) around 650 CO_2 equivalent.

RCP2.6: Radiative force peak at approx. 3 W m^{-2} before year 2100 and then declines, and carbon concentration (p.p.m) peak at approx. 490 CO_2 equivalent.

In fact, the representative characteristic in the new scenario allows the consideration of all greenhouse gases and land use change and further corresponding them to the four chosen concentration levels specified above. Thus, the pathways projections are consistent with the social and demographics. In sum, to determine the future projections for Malawi energy system, potential pathways in form of scenarios will be ideal. In fact, the RCPs have been used as the basis for simulations in integrated assessment modelling to produce projections of the magnitude and pattern of climate change over the 21st century (Tylor et al., 2012 and Ebi et al., 2014). The scenarios within the LEAP system incorporates the radiative forces and carbon concentrations of these RCPs.

4. Analysis

4.1. LEAP System

The model presented in this study is based on national level energy use behaviours. Malawi household energy sector can be divided into urban and rural. However, data from the Ministry of Energy (NSO, 2013; GoM, 2011b; 2009; and 2017c) shows similar behaviors in energy use and demand in both urban and rural settings. For instance, in both areas charcoal and firewood are used in high quantity for cooking in comparison to electricity (Openshaw, 2009). As such, the analysis has grouped the household as one sector. The other demand sectors are service, industry, agriculture, and transport (see figure 2.2 above).

The LEAP system has simulated data for the period 2008 to 2063. The main model comprised of population size, economic growth, energy adjustment (improvements in efficiency, affordability), structural strategic changes and adoption of renewable energies. Using the built in environmental loading tool that shows the effect of a given technology, the study calibrates long term emissions figures and emission levels from different energy sources to develop optimized mitigation options.

4.2. Model Building

The main objective of this analysis is to address the research questions raised in chapter one. In that regard, three scenarios outlined in section 4.2.1 below, have been developed for energy demand and supply projections, and negative environmental effects simulations. These scenarios are based on baseline socio-economic indicators of the country outlines in table 4.1 below.

Table 4.1. Socio-Economic Indicators 2008 to 2015

Item	Unit	2008	2015
Population	Million	13.077	15.334
Average population growth rate	% per annum	2.5	
Average Capita/Household	Capita	4.56	4.42
Households	Million	2.89	
GDP	Billion US\$	3.499	5.618
Average Total GDP Growth Rate	% per annum	3.5	

4.2.1. Scenario Building

Scenario 1. Business as Usual (Reference Scenario)

The Business As Usual (BAU) or Reference (REF) scenario is the baseline quantification of all existing policies prior to the base year (2000 to 2008). Any future policies beyond a base year, are excluded. The main components of the analysis are the trends in economic growth, population growth, industrialization, urbanization, and energy. Table 4.1 above offers a summary of the key assumptions for the BAU or REF scenario.

Table 4.2 and Figure 4.1 presents the total electricity consumption time series from 2001 to 2008 (ESCOM, 2011). These trends represent 4.4% average annual growth rate.

Table 4.2. Electricity Consumption Time Series 2001 to 2008
Source: ESCOM, 2011

Year	2001	2002	2003	2004	2005	2006	2007	2008
Total Electricity Consumption (kTOE)	77.78	76.03	80.63	85.38	90.73	92.58	95.42	104.8

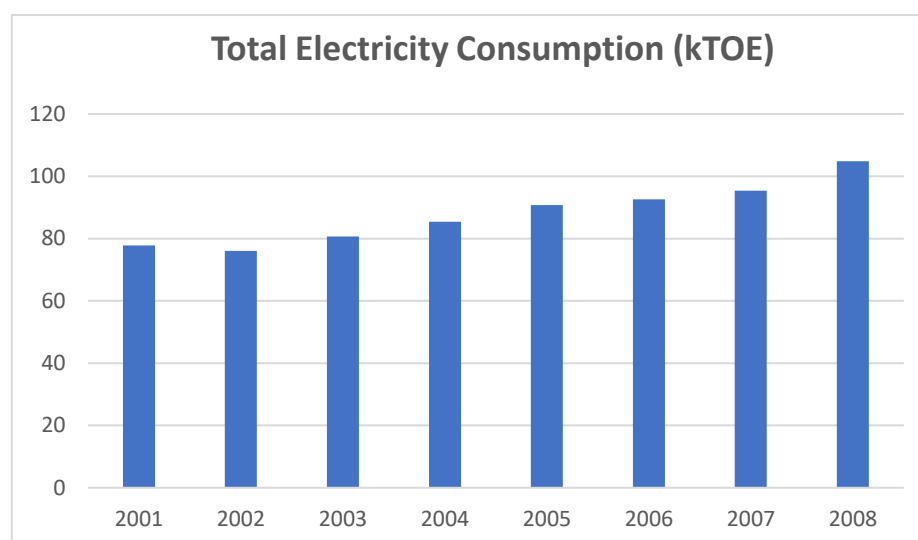


Fig 4.1 Total Electricity Consumption

Scenario 2. Malawi National Energy Policy (MwNEP)

This scenario is based on the background information from sections 1.2 and 1.3 on Malawi Energy Policy goals. Based on the Malawi National Energy policy, Annual Economic report of 2009, the Housing Census 2008, vision 2020, the third Integrated Household Surveys 2010 – 2011, and observations from trend of other neighboring countries in energy consumption patterns like; Zambia, Zimbabwe and Tanzania (EIA, 2018). This scenario can also be regarded as medium accelerated growth with an assumption that some of the policy goals and targets set in the reference scenario may not be achieved.

Table 4.3. MwNEP Scenario Assumptions

Economic assumptions:	Energy and Demographic Assumptions
Income per capita is aimed at reaching the level of middle-income countries by 2030. However, since the assumption is that it may not be ideal then the assumption is that income per capita will double by 2030.	With the transformation from low income to a middle-income country, the share of settlement with urban patterns of energy demand will increase considerably leading to decrease in biomass energy.
Average annual GDP growth will be at 5% by 2020 and beyond.	The National Energy policy aims at increasing the electricity (Hydro-power and other renewables) energy in the energy mix. The core emphasis is transforming the energy systems from overly dependent on biomass (93%) into modern energy (electricity).

Share (GDP) of industry (mining and manufacturing) will increase considerably surpassing the agriculture sector.	One major adjustment on the electricity generation Side is the Government energy plan of constructing the 300MW Kammwamba Coal fired thermal power from the year 2020. Generation of 20MW from diesel generators by the Electricity Generation Company (EGENCO) (EGENCO, 2018)
	Population growth rate will go down following past long-term trend, as the Malawi
	According to ESCOM's sales and generation statistics reports, total system losses for the base year (2008) are 21% of energy sent out to transmission grid. Of these, 7% are transmission losses while 14% are distribution losses (GoM, 2011b).

Scenario 3. Sustainable Energy Access Scenario

In the African context, SE4All and SDG7 focuses on investing in doubling the renewable energy in the global energy mix. Furthermore, and improving the energy efficiency which will ensure development of local resources necessary to increase access to electricity and sustainable energy on the continent (UN, 2011, UN,2015, and Ouedraogo, 2017).

The Sustainable Energy Access scenario adopts the UN's three key energy objectives to be achieved by 2030:

Ensure universal access to modern energy services: The focus for this objective is to achieve 100% urban electrification and 95% Rural. Additionally, enhance clean cooking for all households (UN, 2011). The assumptions have been pegged to an average of 98.5% household electrification by 2030.

Double the global rate of improvement in energy efficiency: This is aimed at doubling the rate of energy efficiency from the current levels to achieving the SE4All objective of annual 2.6% drop in energy intensity between 2010 to 2030 and beyond for the case of this study.

Double the share of renewable energy in the global energy mix: All current shares of renewables are assumed to have doubled by 2030 and based on this trend the scenario will project further processes till 2063.

5. Results and Discussions

5.1. Energy Demand Trends

To begin with, in the base year the energy demand was at 4,125 kTOE and the household sector dominated the energy demand at 3,455.7 kTOE with biomass as the highest energy demand for household at 3,400 kTOE. The total energy demand projection results shows an increase to 17,630 and 18,359 kTOE in 2063 for Malawi Policy Scenario and SE4ALL Scenario respectively. Table 5.1 below summarizes the energy demand projections between 2008 and 2063 including the SDG target year of 2030. The energy balance tables A9.4 to A9.6 and sankey energy balance diagrams in figures A9.5 to A9.7 in the appendix gives a detailed data visualization for the energy balance analysis results from the LEAP system on all the three scenarios in this study.

Table 5.5 Final Fuel Demand

Units: Thousand Tonnes of Oil Equivalents					
Scenarios	2008	2015	2030	2060	2063
Malawi Energy Policy	4,124.8	4,735.9	7,054.1	16,205.0	17,630.3
Reference	4,124.8	4,520.9	5,760.4	13,192.7	14,862.3
Sustainable Energy 4 All	4,124.8	4,962.5	7,405.2	16,851.4	18,359.7

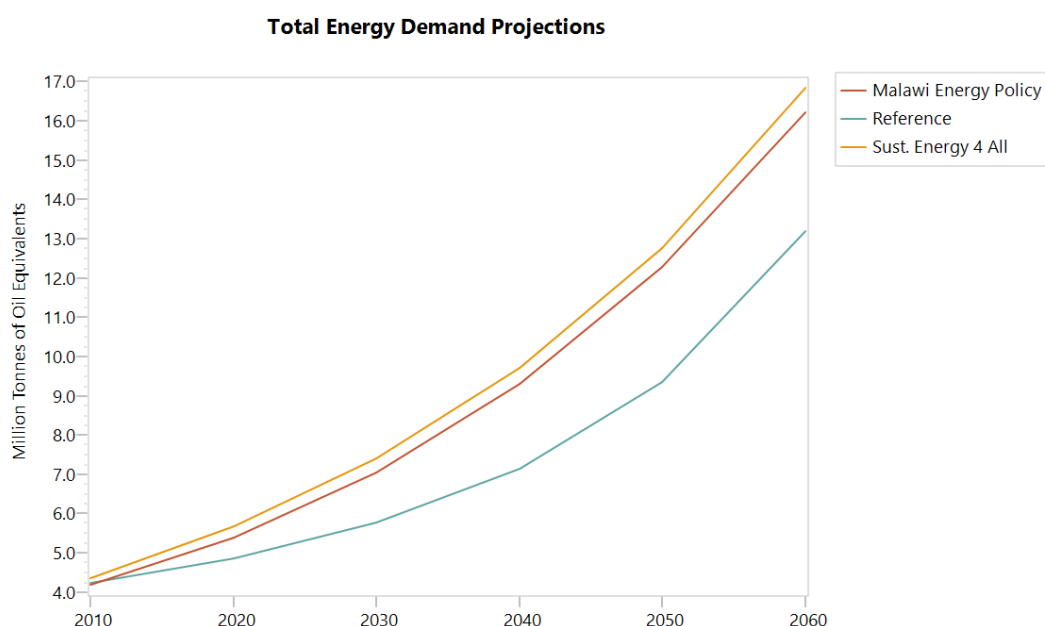


Fig. 5.1 Total Energy Demand Projections

The projections established in the MwNEP Scenario analysis reflected a 1.7% increase in energy demand by 2030 and approximately four-times of the base year by 2063. Similar trends have been observed in the SE4ALL scenario as represented in figure 5.1. This is partly influenced by the SE4ALL objectives that have been incorporated in Malawi energy strategies (GoM 2011b). The MwNEP scenario energy demands are slightly lower than the SE4ALL

scenario projections. These results were expected as the SE4ALL policy are in development stage in Malawi as such parallel trends to National plans should be foreseen (AfDB, 2017).

Unlike in the SE4ALL scenario, where the modern energy mix supplants the traditional fuel (biomass) source in the energy mix, the MwNEP scenario, shows a slow but gradual increase in modern energy lower than the SE4ALL. However, in both scenarios the Biomass energy demands are around the same growth paths with 4,733.2 and 3,176.8 kTOE of biomass demand in 2030 and 4,696.8 kTOE 5,280 kTOE in 2063, for SE4ALL and MwNEP scenarios respectively (see figures 5.2 and 5.3). This shows that the Biomass energy will remain an important energy system in the economy across different scenarios. As such investing in sustainable biomass energy can be deemed as another option while at the same time building the modern energy mix.

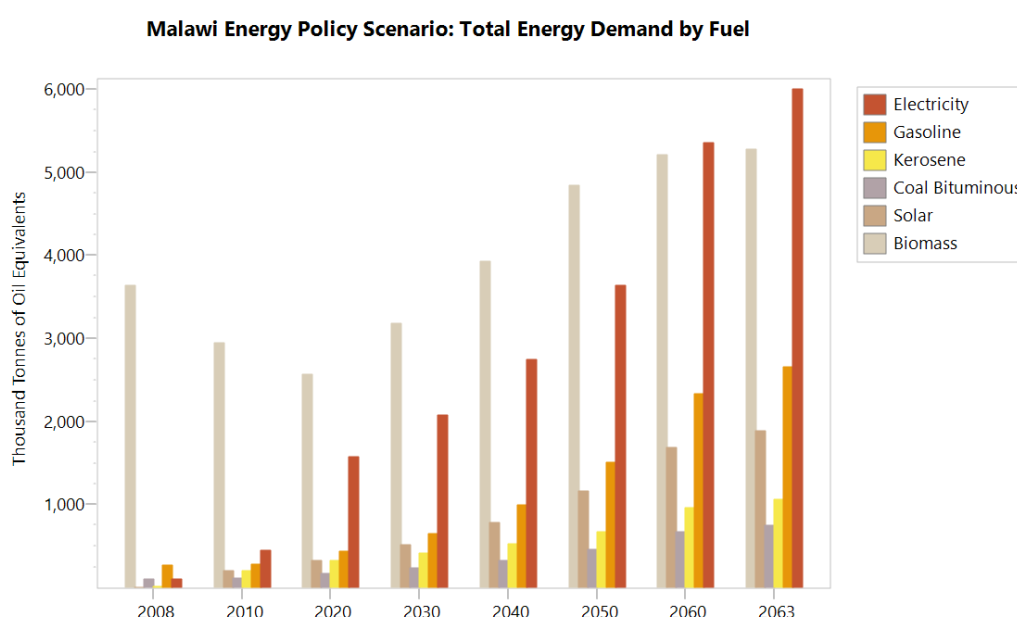


Fig. 5.2. MwNEP Scenario Total Energy Demand

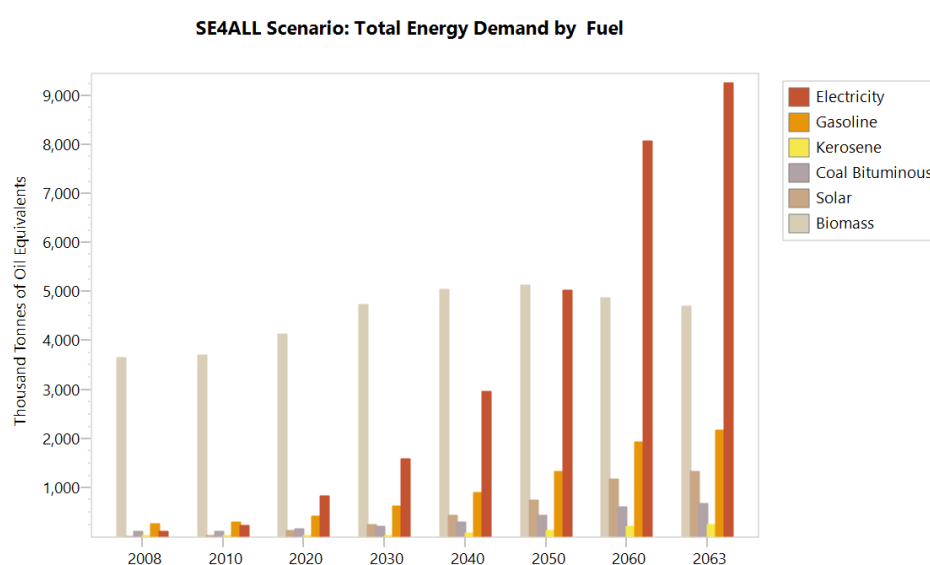


Fig. 5.3 SE4ALL Scenario Total Energy Demand

Furthermore, the analysis on final energy demand by fuel source reflects an increase development in modern energy in the energy mix in all the scenarios over the course of the years. Figure 5.4 and 5.5 summarizes the yearly energy demands for MwNEP and SE4ALL scenarios respectively. This is in line with indicator 7.1 in the SGD 7 which is also reflected in the energy strategies of Malawi. However, we will later see how this increased in modern energy is far from reaching the 2030 goal. Going further, comparing individual energy sources in each scenario, indicates that the SE4ALL Scenario has sharp increase in electricity to 1,587 kTOE in 2030 and to 9,257 by 2063 and a decrease in biomass demand. This increase in electricity demand is comparable to the 2.6% energy intensity drop stipulated in the SE4ALL scenario and the per capita income growth as Malawi strides towards a middle-income country. In contrast, the MwNEP scenario show an increase in demand for both biomass and electricity, where electricity is slightly higher. Thus, the biomass demand will not subside as is the case in SE4ALL scenario. This entails a need to push for energy intensity decrease through efficient energy infrastructure development to drive lower cost of energy conversion.

In both MwNEP and SE4ALL scenarios the energy balance of fuels (energy source) and annual growth rates of modern fuels is not enough to meet the SDG 7 targets by 2030. A Biomass to modern energy fuel ratio analysis indicate that by 2030 biomass will continues to dominate the energy share by 55% and 72% in MwNEP and SE4ALL scenarios respectively.

5.2. Transformation of Energy Supply

Electricity Supply

The hydro-power electricity generation of 360MW at 80.2% and 98% installed capacity and electricity generation levels respectively is expected to be supplemented by the 300 MW planned Kammwamba coal-power plant by 2020 and 72 MW from solar plant, hydro-power plant rehabilitation and diesel generators (GoM, 2010 and GoM, 2017d). This is an agenda put forward by the Malawi government to accelerate the transition to modern energy mix (GoM, 2017d). However, by 2020 the electricity demand based on the LEAP projections indicate an approximately energy demand of 18,299.7, and 9,691.8 GWh in MwNEP, and SE4ALL scenarios respectively. Thus, the electricity demand in REF scenario will not be met with such generation capacity for the MwNEP, and SE4ALL scenarios. As we will observe later in Kambewa et. al. (2007) and Missanjo et al. (2015) studies on the role of biomass in the energy mix, it can be argued that the electricity energy gap observed here is supplemented by biomass.

Biomass Supply

Kambewa et al. (2007) projected an increase in sustainable biomass energy production if government stricken the implementation of the Biomass Strategy. In figure 5.6, Kambewa et al. (2007) summarizes the biomass energy transition process for sustainable energy scenario. It can be observed that in a Business as Usual (baseline) scenario, the biomass supply which is dominated by wood declines to almost equal the biomass demand. Thus, a scenario implying massive deforestation (Kambewa, 2007) which will push the energy demand on biomass further below beyond 2030 leading to massive environmental degradation. As such the carbon sequestration function of forests will be heavily affected (Missanjo et al. 2015). Contrary, a sustainable biomass energy expects to keep the biomass energy supply at 300 million tonnes.

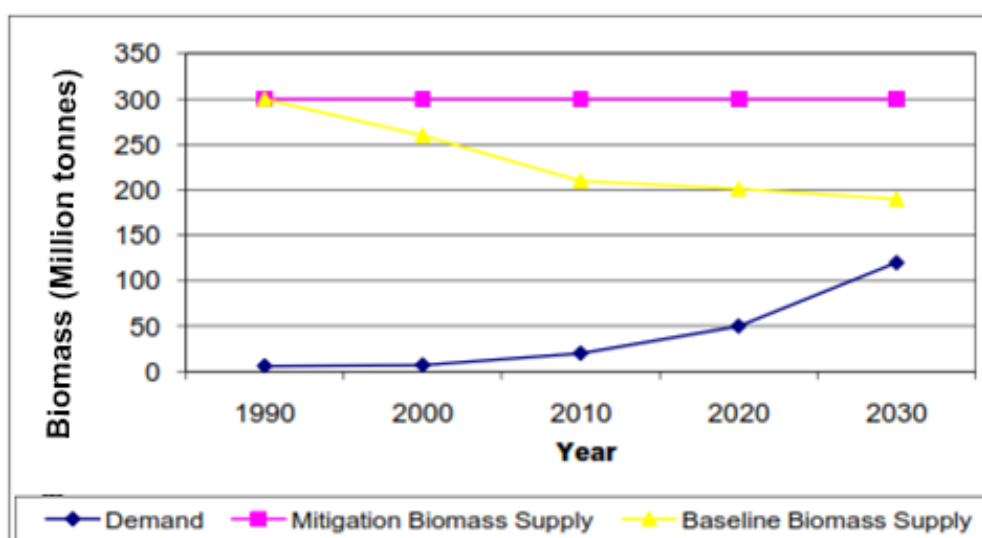


Fig. 5.4 Biomass Supply Projections
Source: Kambewa et al. 2007

To sum up, incorporating other renewable energy sources has the potential to increase accessibility of renewable energy. With the energy potential Malawi has, solar energy can fall among the energy options to be capitalized. This is because of its lower demand for infrastructure in off grid, unlike in grid electricity which require specific housing requirements that most rural Malawian household do not have. A solar mapping modeling study by the World Bank showed that the central and northern regions present high solar radiations, which correspond to high Global Horizontal Irradiation (GHI) used for solar energy conversion. These regions average daily totals reach 6.0 kWh/m² (yearly total 2190 kWh/m²) and more (Suri, et al. 2015). The GHI “is used as a reference value for comparing geographical conditions related to Photovoltaic (PV) electricity systems, ignoring possible modifications, given by choice of PV system components and the configuration of a module field” (Suri et al. 2015).

5.3. Greenhouse Gases (GHGs) Emissions

The GHGs emissions have been simulated in the LEAP system using the IPCC tier 1 default emission factors which are pre-installed in the system. The summary of GHGs emission levels for Malawi in figure 5.5 shows that in the Malawi National Energy policy scenario there is a high GHGs emission level in MTCO₂ equivalent, whereas the SE4ALL scenario has the least of the two scenarios. This is expected as the former is geared towards minimizing the anthropogenic climate change.

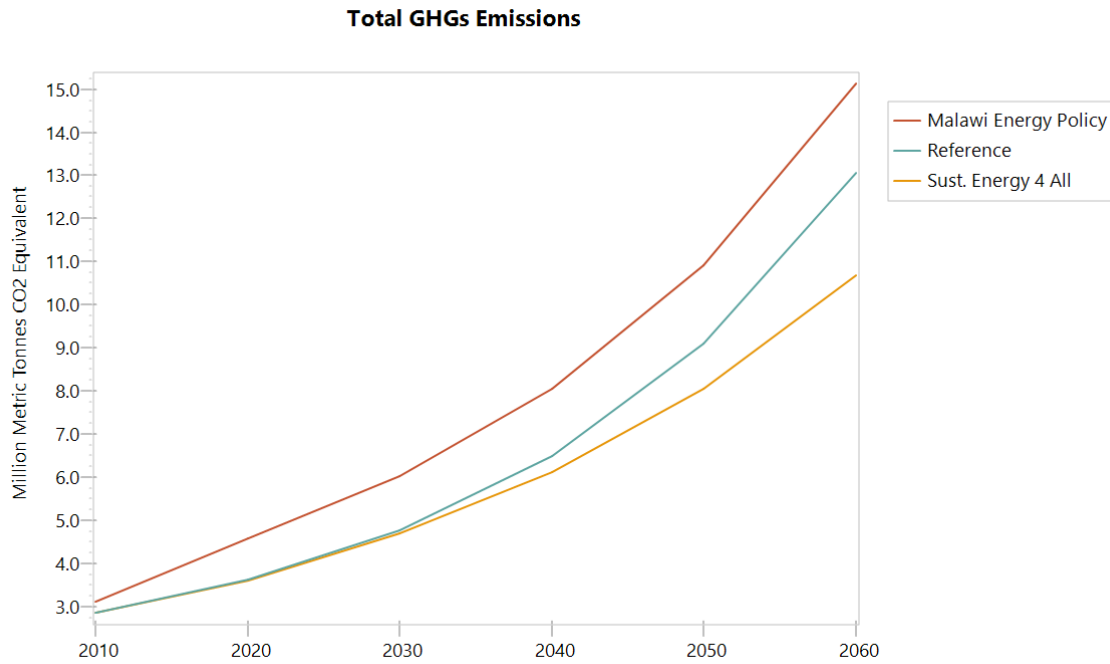


Fig. 5.5 Total GHGs Emissions projected from 2010 to 2060 within 3 different scenarios

In all scenarios summarized in figures 5.6 to 5.8, we observe high GHG emission levels derived from Gasoline which around up to about 7,000 (MTCO₂ Equivalent) by 2063. By 2030, we expect similar emissions from biomass and gasoline as the country shifts from heavy consumption of biomass to use of modern energy (see figures 5.6 to 5.8).

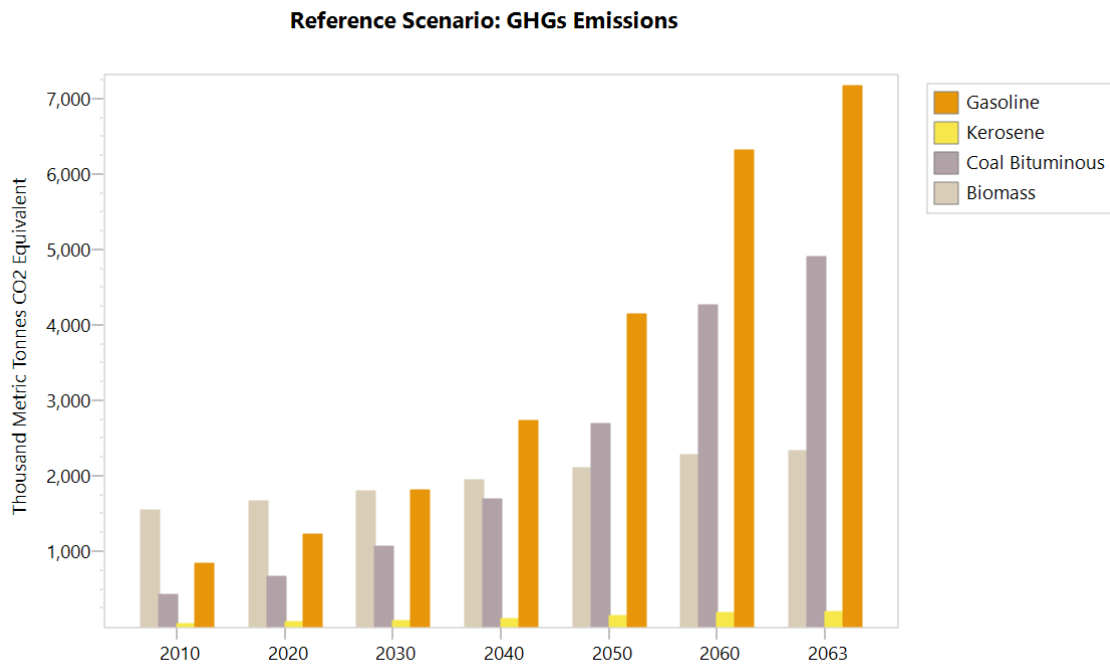


Fig. 5.6 REF Scenario GHGs Emission by Fuel for 2010-2063

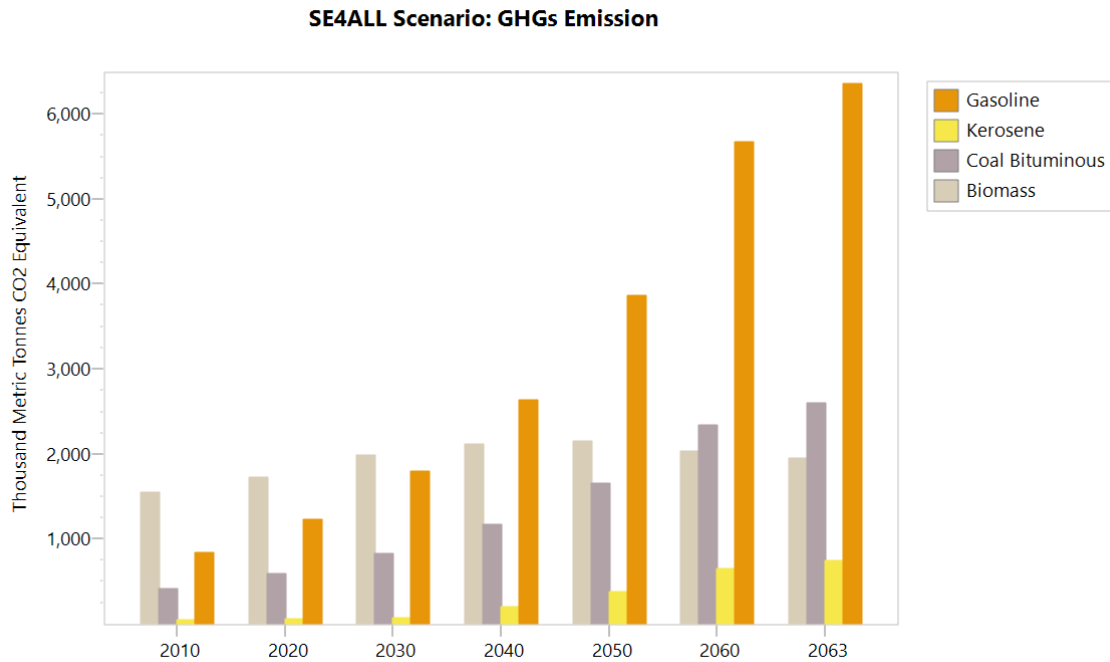


Fig. 5.7 SE4ALL Scenario GHGs Emission by Fuel for 2010-2063

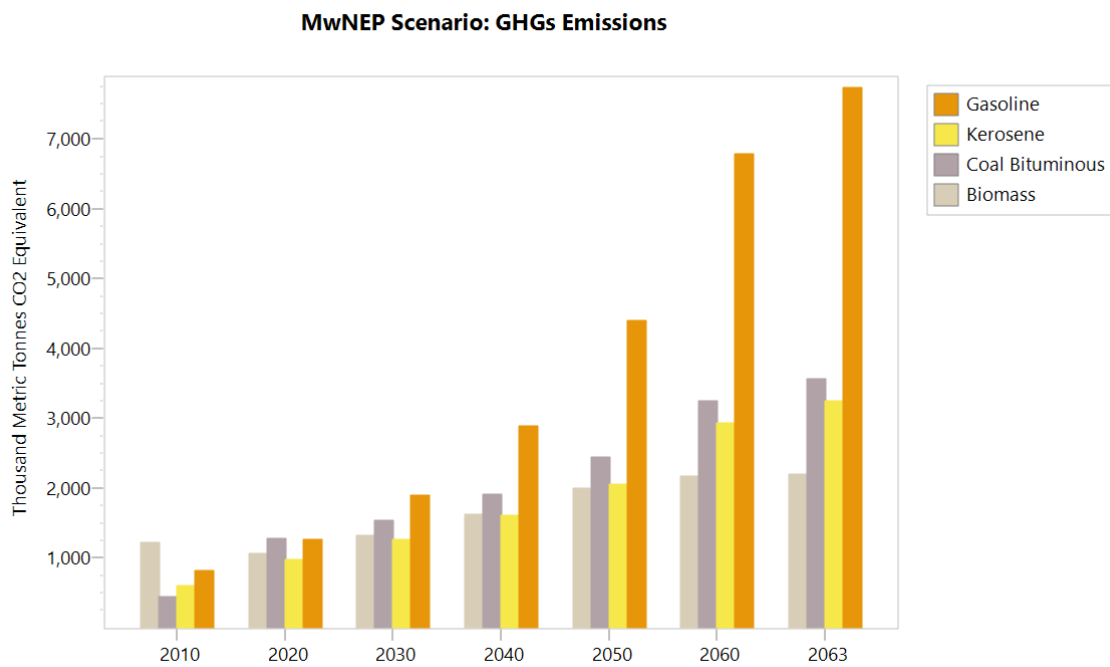


Fig. 5.8 MwNEP Scenario GHGs Emission by Fuel for 2010-2063

The socio-economic assumptions summarized in table 4.3 above contributes to creating a favorable environment for Malawi to transition into a middle-income country. As the country moves towards a middle-income level, the demand for transport energy and industrial locomotive energy generation systems, will increase (GoM, 2011a, Ministry of Transport and Public Infrastructure, 2017 and ENGECO, 2018). This increase corresponds to the annual increase in vehicles of 12% (Ministry of Transport and Public Infrastructure, 2017) which

creates shocks in transport GHG emissions from petroleum (gasoline) products. In addition, the petroleum fuels consumption will increase with increasing demand from diesel electricity generation proposed by the Electricity Generation Company (EGENCO), and water pumping in agricultural irrigation farming. Regardless, the transport system remains as a high contributor of GHGs emission through motor fuels in both medium and long-run terms. Fig. 5.9 to 5.11 gives an overview of GHG emissions by sector for 2015, SDG target year 2030 and 2060. It is observed from the projection that the transport sector's GHG emissions from petroleum in the Malawi Policy scenario (MwEP) will increase exponentially above other energy sources in the specified five sectors.

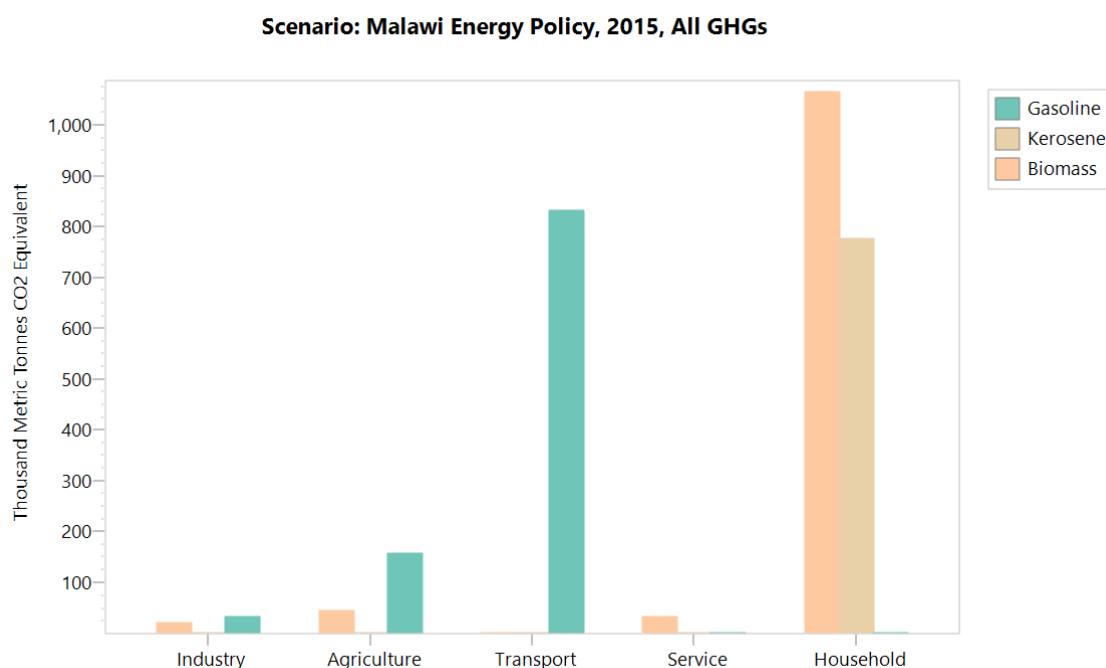


Fig. 5.9. GHG Emissions by Sector/Branch for 2015

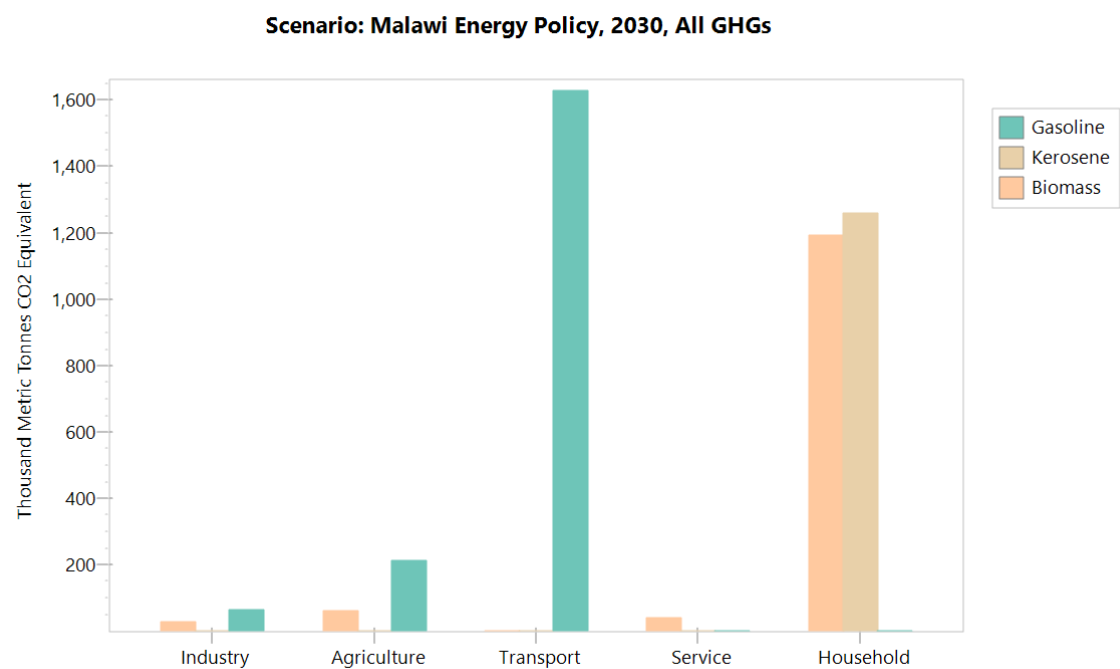


Fig. 5.10. GHG Emissions by Sector/Branch for 2030

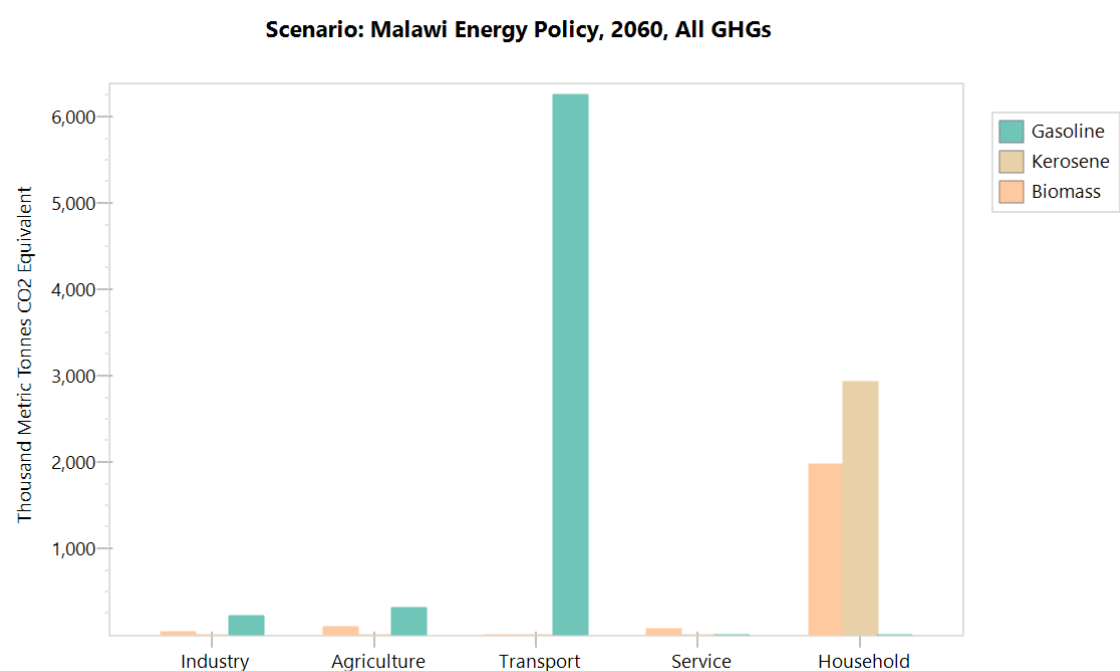


Fig. 5.11. GHG Emissions by Sector/Branch for 2060

5.4. Discussion

The demand for biomass in the MwNEP scenario will continue to grow but just below that of modern energy (electricity) by 2060. However, the household sector will continue to dominate the energy demand. The increasing demand for biomass will affect the carbon sequestration leading to high carbon accumulations as more woodland is cleared for energy and farming. In the medium term to the SDGs target year of 2030, the projections across all the three scenarios show the dominance of biomass energy use. The household sector in MwNEP scenario is projected to have the majority share of 120.4 MGJ for biomass energy demand (see table A9.2 in Appendix). This represents a high continuum of biomass energy demand which will inevitably reduce the wood cover and lower the carbon sequestration process if no strict measures are put in place to resolve this problem. However, introducing sustainable forestry like the Swedish Sustainable forestry (Helander, 2015) can enhance afforestation and reforestation. It can be contended that Sustainable forestry will speed up the wood recovery process for carbon sequestration and create a buffer stock of biomass (wood and forestry residual) for energy use (Openshaw, 2009; Helander, 2015).

Furthermore, as the per capita income increases, we expect more middle-income households to transform their energy consumption behavior and incorporate frequent travelling and private vehicles in their livelihood at a 12% growth rate (Ministry of Transport and Public Infrastructure, 2017). This will inevitably lead to an increase in demand for gasoline. Consequently, the GHG emission levels derived from transport and household are expected to continue increasing. Likewise, in the energy generation sector, the proposed 20 MW diesel generators by ENGECO and the construction of the 300MW coal-powered power plant will add GHG emissions to the energy sector (EGENCO, 2018). Therefore, the country will need to focus on diversifying the energy sector to shift the demand from biomass and fossil energy to sustainable modern energy with extremely low levels of carbon footprint. The transformation of the public transport to run on renewables while delivering excellent services similar to the private transportation system, will encourage use of public transport over private vehicles. Thus, encouraging the path towards GHG emission reduction from the transportation sector.

There is a high investment on energy conservation at global level in the implementation process of the Sustainable Energy for All (United Nations, 2011; Mandelli, 2014). However, in the Malawi's energy policy (MwNEP scenario) the focus is at meeting the minimum level of modern energy demand for the population. Hence, there is a priority at extending electricity generation than efficiency. This corresponds to Mandelli (2014) and Ouendrago's (2017) studies in which they observed that the global community including UN's SE4ALL, invest towards improving energy efficiency which is contrary to objectives of most Sub-Saharan African countries. As a result, the energy policy implementation output from Sub-Saharan African countries including Malawi show contrary results.

The SE4ALL scenario indicates increase in energy demand over MwNEP and REF scenarios. Contrary to the MwNEP scenario, the SE4ALL scenario has less GHG emission projections which is ideal to its objectives. The low emissions count can be attributed to the doubling of renewables in the energy mix. Thus, adopting options within SE4ALL into national energy policies will be instrumental to adjusting the country's current plans. However, these options will need to be localized to meet the areas of need within the country.

Despite these projections in all scenarios the SDG7 will not be achieved by the year 2030, as shown in Figure 5.2 and 5.3 above, where the demand for biomass remains dominant in both scenarios. Among the reasons include the low modern energy generation within the country to meet the demand. However, there is an expectation of closing the gap beyond 2060. This

analysis show room to focus on other energy supply systems including introduction of rural based mini-grid stations for hydro and solar that will speed the process of covering the high rural demand that currently depend on biomass and kerosene forms of energy. In addition, importing grid energy from neighboring countries as an addition to local energy sources (hydro, solar and wind). In energy mapping study for Malawi, Suri et al. (2015) observed that photovoltaic technology (PV) power production vary between 3.6 kWh/kWp (equals to yearly total of about 1315 kWh/kWp) and 4.8 kWh/kWp (about 1750 kWh/kWp yearly) which can contribute to the energy gap.

Furthermore, Malawi can leverage on it available energy generation by reducing the energy losses during transmission and distribution in addition to off grid solar panel electricity for its 85% rural households which are not designed to accommodate grid electricity. Similar recommendations have been emphasized both in Mandelli (2014) and Ouedraogo, (2017b) studies. They both content that most Sub-Saharan African countries record high transmission and distribution energy losses due to aging and inefficient power generation plants in addition to poor infrastructure for large scale modern electrification in their rural areas. It is therefore critical to align the policies towards renewable energy production, efficient energy transmission and distribution to improve liability of the system and increase energy access. Thus, plans aimed at constant maintenances and replacement of energy generation and transmission infrastructure has the capability to reduce the energy loss in transmission. Therefore, an ideal energy system should encourage grid and off-grid sustainable energy generation while reducing power loss through transmission.

Overall, a national wide investment in enhancing the energy planning capacity for the energy planners will be ideal at increasing the country's ability to anticipate and respond to energy shocks and opportunities. Bizilian et al., (2012) observed that encouraging government commitment to energy planning lead to comprehensive energy systems in relation to energy related policy development and investment decisions. In this case, more effective energy planning will leverage on energy demand projection on this and similar studies and correspond them to different energy sources specified in energy mapping by Suri et al. (2015). Currently, MAREP project has proved tremendous developments at electrifying rural trading centers from 45 in 2005 to 182 centers in 2010 (GoM, 2012). However, it forgoes the rural community households that demand off-grid technology.

Suri et al. (2015) contends that PV will dominate the renewable energy in Malawi and is best supplement to the hydro-electricity. For Malawi two PV systems types are relevant, the mini grid-connected power plants like the case of Sitolo solar mini-grid in Mchinji and the off-grid PV systems (Suri et al. 2015). Acknowledging the financial barriers of the rural population to access the PVs, proper facilities should be put in place to encourage accessibility and affordability of the technology. For instance, integrated renewable energy financing in form of subsidies, and flexible payment for PVs that allows poor household to pay the total cost in instalments without interest will accommodate their financial capabilities. Accessibility can be enhanced by increasing information distribution about the technologies and increasing PV availability to local traders where people can easily access. Furthermore, min-hydro plants have been envisaged as potential clean energy scale-up options (EGENCO. 2018a). Some examples include, the kavuzi in Mzimba, and Mulanje Electricity Generation Agency (MEGA).

6. Conclusions

Energy is regarded as the core pillar in building the national economy (GoM, 2017). Therefore, to enhance the energy sector there is need for better policy planning that will ensure secure energy now and in the future. To account for this long-term energy planning, this study used the LEAP system to analyse three policy pathways (REF, MwNEP, and SE4ALL) and forecast potential energy pathways for Malawi from 2008 to 2063. LEAP system uses demographic, economic and energy intensities to derive energy demand trends and environmental impacts. The SE4ALL scenario has been analysed together with MwNEP and REF scenario because of its global considerations as a sustainable energy plan. Currently, Malawi energy policy is old (2003) and the country is transitioning from the current long-term economic development policy called vision 2020 which is expected to end in the year 2020. Consequently, these analyses are important for any long-term energy and economic development planning.

The LEAP modelling framework analyses show a tremendous impact of national and international economics development policies on energy demands and environmental GHG emissions for Malawi. The scenarios analysis of MwNEP and SE4ALL generates a similar exponential energy demand growth projection between 2008 and 2063. In all the scenarios, more pressure is exerted on small local energy supply sector creating an energy demand gap between available and demanded energy. Furthermore, economic development is increasing the energy demand and the country responds by generating unsustainable energy from fossil fuel. As a result, there is a surge in GHGs emissions with increasing energy consumption demand. Despite the similarities in energy demand for all the scenarios, MwNEP focuses on extending and increasing its generation and supply, while the SE4ALL emphasizes on doubling renewables and enhancing the energy transmission efficiency. This is reflected by low energy loss in transmission for SE4ALL scenario in comparison to MwNEP scenario.

The energy trends in both MwNEP and SE4ALL indicate that the long-term demand for electricity will be higher than the available supply which will inevitably affects the 2030 goal of universal access to modern energy set by SDG 7. The hydroelectric power generation is often constrained to sufficiently meet the electricity energy demand by lack of infrastructures, droughts and decrease in water levels (USAID, 2017). Similarly, the demand for biomass suppresses electricity demand in both scenarios, thereby affecting the access to clean energy (electricity) by 2030. Consequently, there is a demand for more sustainable energy generation systems to meet the growing electricity demand.

On the environmental side, the MwNEP scenario portrays high GHGs emissions in its energy mix, because of continued reliance on biomass at household level and growing fossil fuel demand in the transport and energy generation sectors. The country can leverage on the SE4ALL strategies that focuses on total clean energy sources. However, to abate the high carbon footprint in the energy sector, sustainable energy transformations are required in sectors with high carbon footprint from energy use. For instance, at household level, sustainable forestry and agroforestry has the potential of sustainably supplying fuelwood without having ripple-effects in the carbon sequestration and other environmental sectors in the medium term while the country transitions to a total use of clean energy. In the transport and energy generation sectors, developing energy plans that improve public transport infrastructure, controlling high emitting vehicles, and transitioning from fossil electricity generation will have a high impact in reducing carbon footprint. Furthermore, taking advantage of the existing ethanol biofuel sector and advance its development for use in the gasoline fuel mix, will have the potential to adjust the demand for biofuel vehicles. Thereby cutting GHG emissions and fossil fuel cost.

Energy financing and investment can be boosted through integrated energy financing approaches that create a conducive energy investment environment and incorporates other sectors to easily invest in energy. Furthermore, pursuing policy plans that aim at reducing the cost of renewable energy while allowing the population to procure renewable energy materials like PVs at a flexible payments method can have potential at scale towards adoption.

Therefore, to achieve sustainable energy in Malawi, there is need for sound integrated sustainable energy planning that considers all the social economic and environmental sectors of the country. The energy that is able to meet the current and future demand without heavy costs on the environment will be deemed sustainable for Malawi. However, to achieve this requires high investments in knowledge and finance on domestic renewable energy resources (like solar, wind, sustainable biomass). In addition, implementing policy plans that aim at electricity efficiency, and regulation fossil fuel use in transport and energy generation. These transformation changes will fuel continued socio-economic development without increasing emissions of GHGs.

6.1. Future studies

To sum up, this study has suggested both introduction and improvements of the energy infrastructure to increase the sustainable energy supply. However, the costs associated with infrastructure development are not part of this research. Therefore, future studies in energy policy planning in relation to costs are required. Furthermore, a disaggregated research that investigates individual sectors in detail will be ideal to understand individual energy processes on demand, supply and conservation. In this study, the missing data on activity level and final energy intensity the study used aggregates from countries with similar energy system as Malawi. As such there remains a gap for an independent study to compile a comprehensive database for energy variables that can easily be used for energy modeling on different systems including LEAP. Lastly, there is a need for a comprehensive GHG emission data from all sectors including transport and energy sector. Although this analysis has been able to project emission levels across sector, a thorough comprehensive study on GHG emission is required to provide up to date data from the current official GHG inventory data of years up to 2010 (GoM, 2011b).

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9. Appendices

9.1. Appendix I: Systematic Literature Review Schema

Table A9.1 Systematic Literature Review Schema

Research Question (RQ)	Source	Literature Search	Screening Literature	Critical Appraisal	Synthesis and Dissemination
RQ 1: How are the energy demand trends in existing Malawi energy policies compatible with sustainable energy as specified in SE4All and the SDG 7? RQ 2: What is the relationships between projected future energy demand and available current energy resource stocks? RQ 3: What alternative energy planning, energy source and supply options can be leveraged to attain a sustainable energy future, and reduce emissions?	Online	Government of Malawi: NSO; Google Search of keywords "Malawi Energy", "Systematic Literature Review of Malawi Energy", etc. UN IRENA. IEA. www.ub.uu.se. https://www.journals.elsevier.com/energy . JSTOR online	RQ 1: Historical and current Energy demand trends for Malawi. Malawi socio-economic and population demographics. Energy consumption behaviour Energy plan for Malawi. SE4ALL and SDG 7 indicators and assumptions. Energy statistics (final intensity, energy mix. RQ 2: Energy statistics required for model build and simulation. Literature on Malawi energy production capacity and demand. RQ3. Energy recommendation studies on Malawi, Sub-Saharan Africa and LDCs. Malawi and Sub-Saharan African studies that used LEAP system	Relevance of the literature and data to the RQ 1, 2 and 3. Assess internal Validity of the Literature or study in minimizing biases. External Validity to generalize the finding of the literature and data statistics into the LEAP system. Appropriateness of the data analysis and presentation of the study. Ethical implication of the intervention that the literature under review evaluated	Narrative Synthesis: Develop summaries of results of the studies describing important characteristics of the included studies. then execute the following three steps: (1) logically organize studies in categories. (2) Analysing findings in each category. (3) Synthesize finding across all categories. Quantitative Synthesis: Synthesizing quantitative data from different studies to meet the LEAP system requirements; Have same Standard Unit. This is followed by the three steps described above.
	Physical Source: Office/Department	Consult/phone contact with Department of Energy affairs in the MNREM. Consult/phone contact with ESCOM officials. Check literature reviews and bibliography of review studies on previous energy studies for Malawi, and Sub-Saharan Africa that were identified in the online sources above.			
	Physical Hand searching in the Library	Uppsala University Library, Nordiska African Institute and SLU Library.			
	Gray (unpublished) Literature	Official but unpublished articles on websites of; the Government of Malawi, UN, and Energy Agencies.			

9.2. Appendix II: Research outline

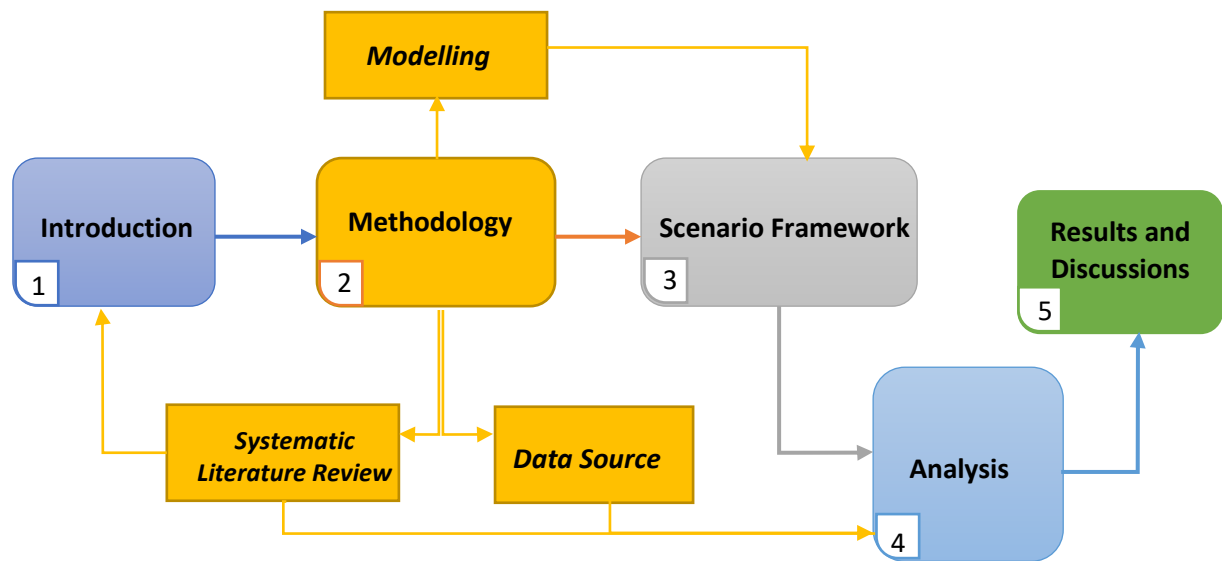


Fig. A9.1. Research Outline

9.3. Appendix III: Transport Emission Diagrams

Reference Scenario: All Fuels, All GHGs

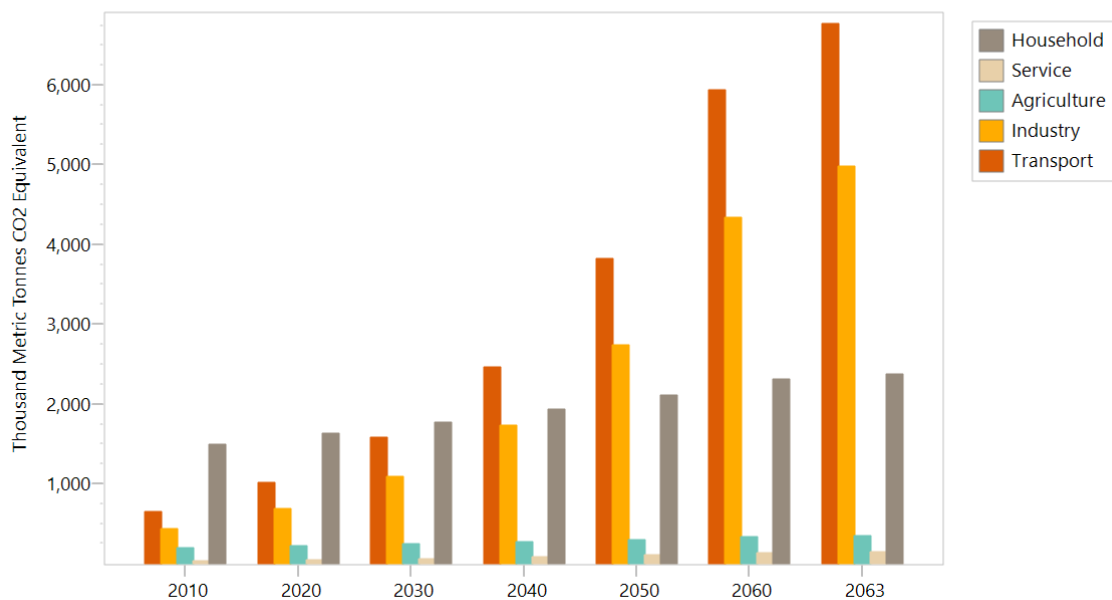


Fig. A9.2. REF Scenario Transport GHG Emissions

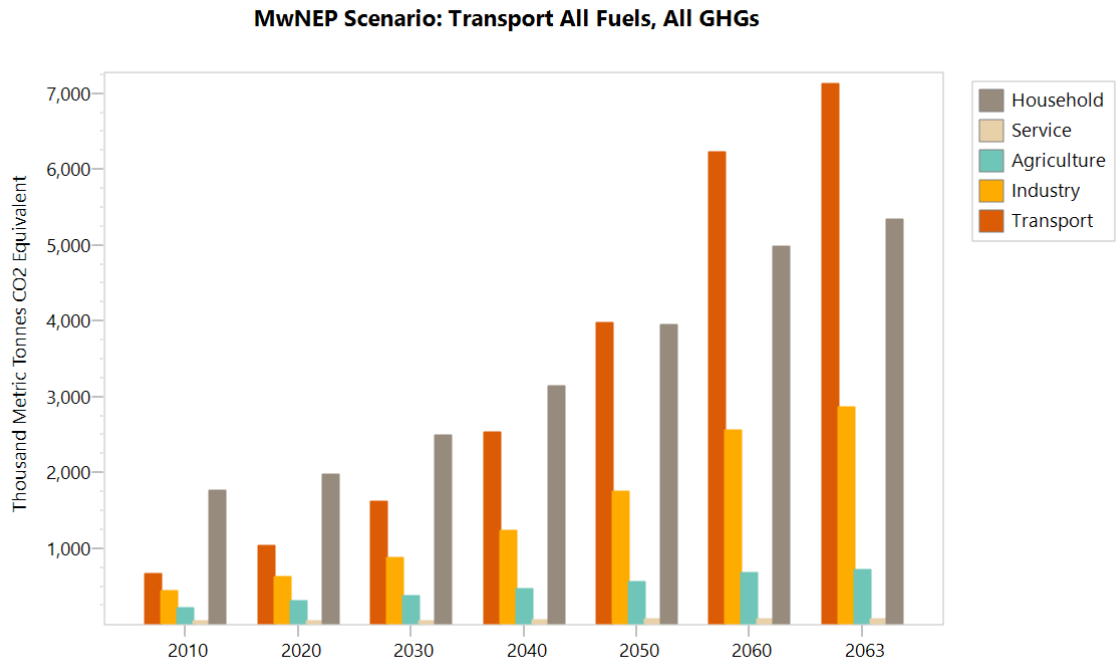


Fig. A9.3. MwNEP Scenario Transport Emissions

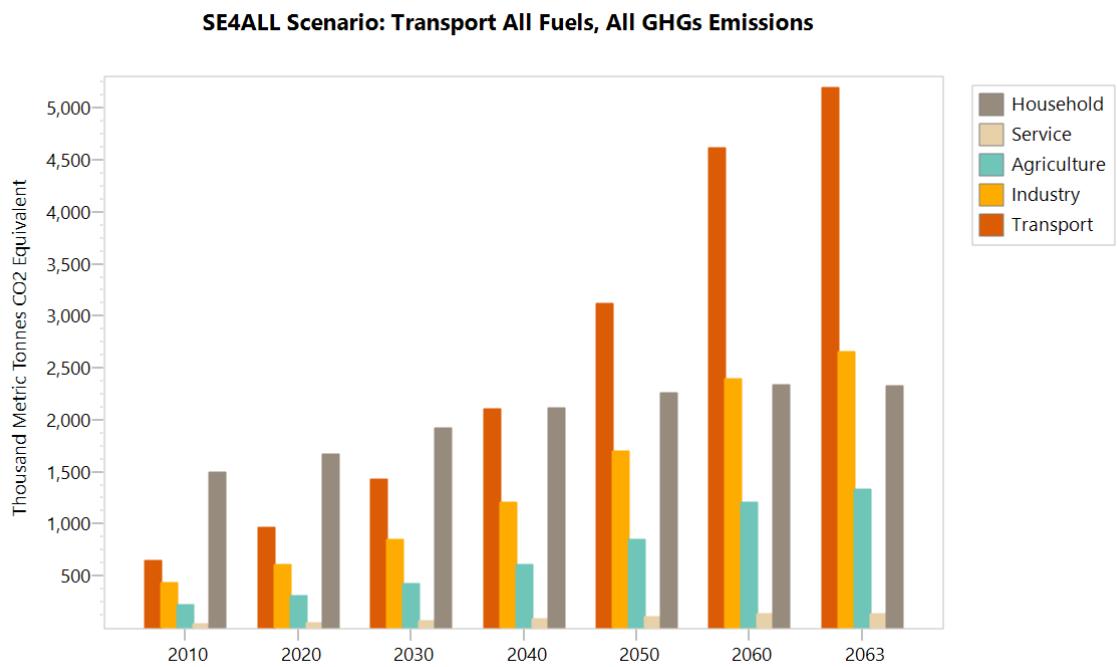


Fig. A9.4. MwNEP Scenario Transport Emissions

9.4. Appendix IV: Energy Demand by sector

Table. A9.2. Energy Demand by Sector 2030

Scenario: Malawi Energy Policy, 2030							
Branch: Demand							
Units: Million Gigajoules							
Branches	Biomass	Solar	Coal Bituminous	Kerosene	Gasoline	Electricity	Total
Household	120.4	20.5	0.2	17.2	-	74.8	233.1
Service	3.9	0.8	0.0	0.0	-	4.4	9.1
Transport	0.0	-	0.0	-	23.3	0.2	23.5
Agriculture	6.0	-	0.9	-	3.0	3.7	13.7
Industry	2.7	-	8.6	-	0.9	3.8	15.9
Total	133.0	21.3	9.7	17.2	27.3	86.8	295.3

Table. A9.3. Energy Demand by Sector 2063

Scenario: Malawi Energy Policy, 2063							
Branch: Demand							
Units: Million Gigajoules							
Branches	Biomass	Solar	Coal Bituminous	Kerosene	Gasoline	Electricity	Total
Household	202.6	74.9	0.6	44.5	-	197.0	519.5
Service	6.5	4.0	-	0.0	-	16.8	27.4
Transport	0.0	-	0.0	-	102.9	0.4	103.3
Agriculture	9.4	-	2.8	-	4.7	22.7	39.5
Industry	2.6	-	27.8	-	3.6	14.3	48.4
Total	221.1	78.9	31.2	44.5	111.2	251.3	738.1

9.5. Appendix V: Energy Balance Tables

Table. A9.4. 2008 Energy Balance Base-Year

Scenario: Reference, Year: 2008, Units: Thousand Tonnes of Oil Equivalent						
	Solid Fuels	Renewables	Biomass	Electricity	Oil Products	Total
Production	-	-	-	-	-	-
Imports	98.6	2.5	3,640.9	104.1	278.7	4,124.8
Exports	-	-	-	-	-	-
Total Primary Supply	98.6	2.5	3,640.9	104.1	278.7	4,124.8
Total Transformation	-	-	-	-	-	-
Household	-	1.3	3,400.6	39.4	14.3	3,455.7
Service	0.9	1.1	80.2	23.4	0.0	105.7
Agriculture	1.5	-	110.4	7.1	49.5	168.4
Industry	96.2	-	49.5	34.1	7.7	187.5
Transport	0.0	-	0.2	0.0	207.1	207.5
Total Demand	98.6	2.5	3,640.9	104.1	278.7	4,124.8

Unmet Requirements	-	-	-	-	0.0	0.0	-
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Table. A9.5. Energy demand Energy Balance table SE4All

Scenario: Reference, Year: 2008, Units: Thousand Tonnes of Oil Equivalent							
	Solid Fuels	Renewables	Biomass	Electricity	Oil Products	Total	
Production	-	-	-	-	-	-	
Imports	98.6	2.5	3,640.9	104.1	278.7	4,124.8	
Exports	-	-	-	-	-	-	
Total Primary Supply	98.6	2.5	3,640.9	104.1	278.7	4,124.8	
Total Transformation	-	-	-	-	-	-	
Household	-	1.3	3,400.6	39.4	14.3	3,455.7	
Service	0.9	1.1	80.2	23.4	0.0	105.7	
Agriculture	1.5	-	110.4	7.1	49.5	168.4	
Industry	96.2	-	49.5	34.1	7.7	187.5	
Transport	0.0	-	0.2	0.0	207.1	207.5	
Total Demand	98.6	2.5	3,640.9	104.1	278.7	4,124.8	
Unmet Requirements	-	-	-	-	0.0	0.0	

Table. A9.6. Energy demand Energy Balance table MwEP 2063

Energy Balance for Area "MwEP_SE4All"							
Scenario: Malawi Energy Policy, Year: 2063, Units: Thousand Tonnes of Oil Equivalent							
	Solid Fuels	Hydropower	Renewables	Biomass	Electricity	Oil Products	Total
Production	-	-	-	-	-	-	-
Imports	985.1	229.2	1,884.2	5,280.0	5,593.3	3,718.8	17,690.6
Exports	-	-	-	-	-	-	-
Total Primary Supply	985.1	229.2	1,884.2	5,280.0	5,593.3	3,718.8	17,690.6
Biomass_Energy	-	-	-	-	-	-	-
Electricity_Generation	241.0	229.2	-	-	410.0	-	60.3
Transmission and Distribution	-	-	-	-	-	-	-
Total Transformation	241.0	229.2	-	-	410.0	-	60.3
Household	13.7	-	1,788.1	4,838.1	4,706.4	1,062.5	12,408.8
Service	-	-	96.0	155.9	402.2	0.5	654.6
Transport	0.0	-	-	0.2	10.0	2,457.5	2,467.7
Agriculture	65.9	-	-	223.7	542.9	111.4	943.8
Industry	664.6	-	-	62.0	341.7	87.1	1,155.4
Total Demand	744.1	-	1,884.2	5,280.0	6,003.3	3,718.8	17,630.3
Unmet Requirements	-	-	-	-	-	-	-

Table. A9.7. Energy demand Energy Balance table SE4All 2063

Energy Balance for Area "MwEP_SE4All"							
Scenario: Sust. Energy 4 All, Year: 2063, Units: Thousand Tonnes of Oil Equivalent							
	Solid Fuels	Hydropower	Renewables	Biomass	Electricity	Oil Products	Total
Production	-	-	-	-	-	-	-
Imports	664.1	867.7	1,327.1	4,696.8	9,039.8	2,414.9	19,010.5
Exports	-	-	-	-	-	-	-
Total Primary Supply	664.1	867.7	1,327.1	4,696.8	9,039.8	2,414.9	19,010.5
Biomass_Energy	-	-	-	-	-	-	-
Electricity_Generation	-	867.7	-	-	216.9	-	650.8
Transmission and Distribution	-	-	-	-	-	-	-
Total Transformation	-	867.7	-	-	216.9	-	650.8
Household	-	-	1,302.8	3,757.6	8,068.1	244.4	13,372.9
Service	16.2	-	24.3	179.9	466.5	0.2	687.2
Transport	0.0	-	-	0.2	175.5	1,790.9	1,966.7
Agriculture	9.8	-	-	685.6	94.3	328.0	1,117.8
Industry	638.1	-	-	73.4	452.3	51.4	1,215.2
Total Demand	664.1	-	1,327.1	4,696.8	9,256.8	2,414.9	18,359.7
Unmet Requirements	-	-	-	-	-	-	0.0

Energy Balance Sankey Diagram 2008

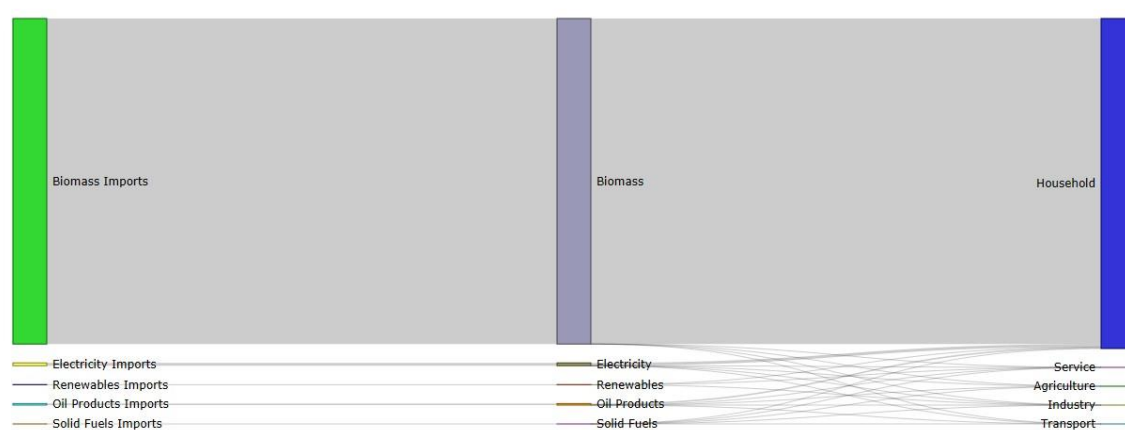


Fig. A9.5. Sankey Diagram 2008 Energy Balance

Reference Scenario 2063 Energy Balance Sankey Diagram

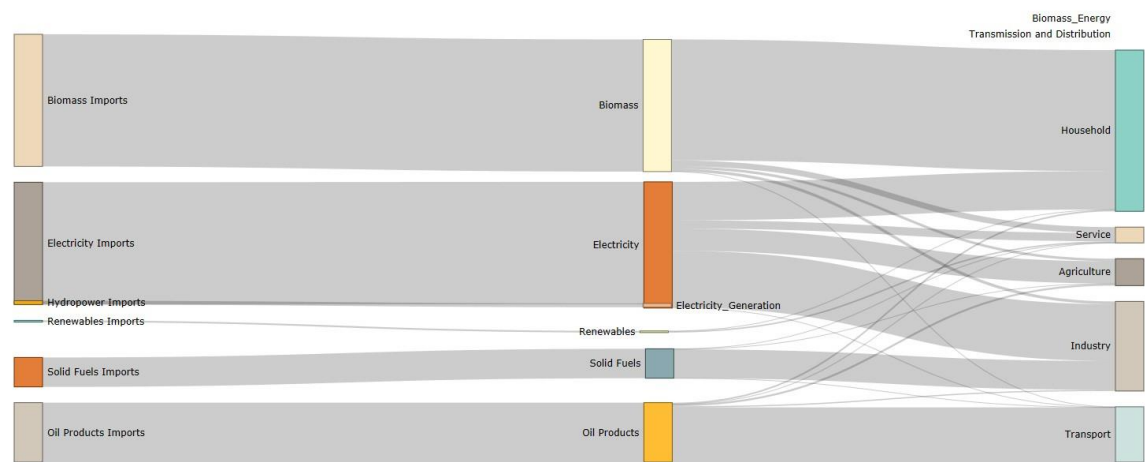


Fig. A9.6. REF 2063 Sankey Diagram Energy Balance

Malawi Policy Scenario 2063 Energy Balance Table

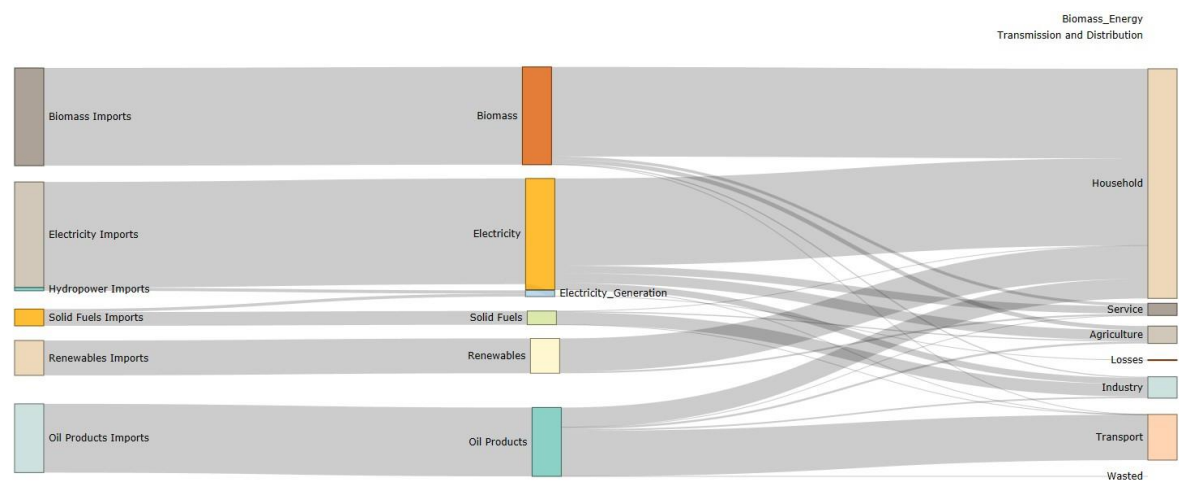


Fig. A9.7. MwNEP 2063 Sankey Diagram

Sustainable Energy for all Scenario 2063 Energy Balance Table

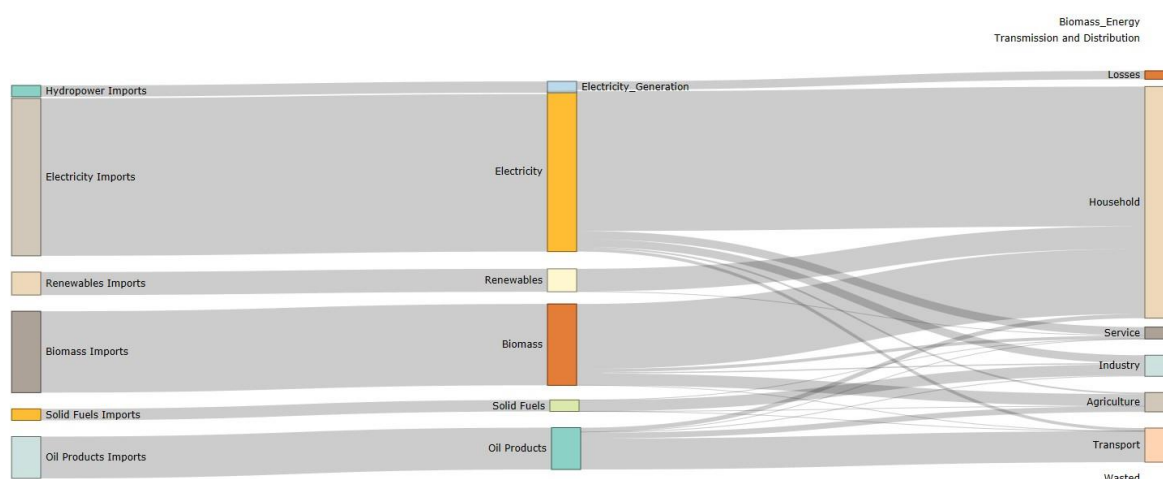


Figure A9.8 SE4ALL Sankey Diagram

9.6. Appendix VI: Energy Measurement and Recalculation Factors

Table A 9.8. Conversion factor for energy units

	GJ	MWh	toe
GJ	1	0.28	0.02
MWh	3.6	1	0.086
TOE	41.9	11.63	1

Table A9.9. Prefixes for energy units

Prefix	Factor	Magnitude	Numerical Value
k	kilo	10 ³	Thousand
M	Mega	10 ⁶	Million
G	Giga	10 ⁹	Billion
T	Tera	10 ¹²	Trillion
P	Peta	10 ¹⁵	Quadrillion

